The Universal Right-Triangle Reflection Sequence: a word-length metric, effective universality, and the lamplighter structure of A396406

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Abstract

For a right triangle T in the plane with unequal rational legs, the group $W(T)$ generated by the three side-reflections acts on the orbit of T, and the orbit-growth sequence u_d^T (the number of triangles at word-distance d) is the subject of OEIS entry A396406. We prove that u_d^T is independent of the shape T through depth 32, and that this is sharp: the $(1,2)$ triangle deviates at depth 33. The common sequence is the orbit growth of the *generic* right triangle, whose abstract reflection group is the group $W = \langle R_0, R_1, R_2 \mid R_i^2 = (R_0R_1)^2 = 1\rangle$. We show that the subgroup of W on which the linear part is a rotation is the wreath product $\mathbb{Z} \wr \mathbb{Z}$ of index 4, so the generic group is virtually $\mathbb{Z} \wr \mathbb{Z}$; the translation lattice is computed exactly as $\mathcal{T} = 2(t-1)\,\mathbb{Z}[t^{\pm 1}]$ via a Mayer–Vietoris/Crowell identification of the translation module with the cyclic $\mathbb{Z}[Q]$-ideal generated by $h = (c-1)(Y-1)$. From the resulting lamp model we derive a closed-form word-length metric whose geodesic length equals a relaxed length plus twice an explicit connectivity penalty. A triangle-independent symbolic normal form reduces every extra collision to the vanishing of a bounded-height integer Laurent polynomial at the rotation number $\zeta_T = (a+bi)/(a-bi)$; Gauss's lemma then gives an effective all-depths universality theorem (no deviation at depth d if $c_T := N(a-bi$, suitably normalized$) > 2d$), and a finite modular certificate closes the gap to depth 32. The natural conjecture that universality holds at all depths is refuted: for every algebraic shape we exhibit an explicit word in $\ker \rho_T$ (a product of conjugated glide-reflection squares), and the first deviation depth n_T is sandwiched by $\max(33, c_T) \le n_T \le 24c_T + 8$; moreover n_T equals exactly half the length of the shortest kernel element — a shortest-vector problem in the ideal $2(t-1)\mu_T\mathbb{Z}[t^{\pm 1}]$ under the lamplighter metric — from which we extract a closed-form three-regime law, $n_T = 6c_T + e_T$, $9c_T - 3e_T$, or $3(c_T + e_T)$ according to the angle of the triangle, certified at $(1,2)$ and validated on thirteen shapes. Finally, the lamplighter transfer yields a q-difference functional equation for the relaxed generating function of A396406; a connectivity-invariance squeeze of the true count between its pure-travel subcount and the relaxed series pins the exact growth rate unconditionally: $\beta_2 = 1.4916177871\ldots$, the reciprocal square root of the smallest positive root of an explicit travel-block equation, which excludes the value 3/2 suggested by finite-depth ratios. The eight length-10 relations and the universality of u_d through depth 22 are formally machine-verified in Lean 4; the depth-32 threshold and the deviation witnesses are certified by exact modular and rational arithmetic. Transcendence of the generating functions over $\mathbb{Q}(x)$, and of the number β_2, lies outside the present scope.

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1 Introduction

Let T be a triangle with vertices in $\mathbb{Q}^2$, and for $i \in \{0,1,2\}$ let R_i be the reflection of $\mathbb{R}^2$ across the line carrying the side opposite vertex V_i. The group $W(T) = \langle R_0, R_1, R_2\rangle \le \mathrm{Isom}(\mathbb{R}^2)$ acts on the orbit of T, and breadth-first search in the Cayley graph relative to $\{R_0, R_1, R_2\}$ stratifies that orbit by word-length. Writing $u_d = u_d^T$ for the number of distinct triangle-images at minimal word-length d, one obtains an integer sequence depending a priori on the shape of T. For right triangles with unequal legs this sequence is OEIS A396406. This paper determines its group-theoretic origin, its exact word-length metric, the precise extent to which it is shape-independent, and its exact exponential growth rate.

The generic group. When T is a non-isosceles right triangle, the two legs meet at a right angle, so $(R_0R_1)^2 = 1$, while by Niven's theorem [8] the remaining two angles are irrational multiples of π and force no further relation among the reflections. The abstract group is therefore the Coxeter group

$$W = \langle R_0, R_1, R_2 \mid R_i^2 = 1 \ (i = 0,1,2),\ (R_0R_1)^2 = 1\rangle,$$

whose Poincaré series is $(1+t)^2/(1-t-t^2)$; consequently the free-group baseline gives $u_d = F_{d+3}$ for $1 \le d \le 9$, until the first affine collision. The shape T enters only through the affine representation $\rho_T\colon W \to \mathrm{Aff}(\mathbb{R}^2)$ realizing the three reflections, and the sequence u_d^T records the layer sizes of $\rho_T(W)$. Our first structural result identifies the kernel of the generic representation. Placing the right angle at the origin and writing $\zeta_T = (a+bi)/(a-bi)$ for the rotation number of the hypotenuse reflection, every group element acts as $z \mapsto \varepsilon\,\zeta_T^k\sigma^\delta(z) + P(\zeta_T)$ for an integer Laurent polynomial P *independent of* T (Lemma, symbolic normal form). The pure-translation subgroup of the resulting symbolic group is computed exactly,

$$\mathcal{T} \;=\; 2(t-1)\,\mathbb{Z}[t^{\pm 1}],$$

by reducing modulo 2 — where the two perpendicular legs collapse onto a single mirror and the group degenerates to the infinite dihedral group, which has no translations. This caps the glide-square lattice $2(t-1)\mathbb{Z}[t^{\pm1}] \subseteq \mathcal{T}$ to an equality (Theorem, translation lattice).

Lamplighter structure and the word-length metric. The translation lattice exhibits the generic group as virtually a classical object: the subgroup $H \le W_{\mathrm{gen}}$ on which the linear part is a pure rotation t^k is normal of index 4 and isomorphic to $\mathbb{Z}[t^{\pm1}] \rtimes_t \mathbb{Z} = \mathbb{Z} \wr \mathbb{Z}$, so every right-triangle side-reflection group with transcendental leg ratio is virtually the wreath product $\mathbb{Z} \wr \mathbb{Z}$ (Theorem, lamplighter structure) — metabelian-by-finite, hence amenable, yet of exponential growth, and not finitely presentable. Independently, the translation module is computed by Mayer–Vietoris for the free product $W = V_4 * \mathbb{Z}/2$ together with Crowell's exact sequence: it is the $\mathbb{Z}[Q]$-module $I_A \cap I_B$, which equals the cyclic ideal generated by the central element $h = (c-1)(Y-1)$ by exactness of the Bass–Serre tree of D_∞ (Lemma, Mayer–Vietoris reduction). The generic kernel is then identified as the relative commutator subgroup $\ker \rho_{\mathrm{gen}} = [T_\rho, T_\rho]$, exhibiting the generic group as the Q-relative metabelianization of $V_4 * \mathbb{Z}/2$ (Corollary, generic kernel). Reading the composition law in the edge basis turns the group into an explicit *lamp model*: a walker on $\mathbb{Z}$ deposits signed lamps on the edges it traverses, with a normal form $(\varepsilon, \delta, k, P)$ cut out by two congruences. From this model we obtain a closed-form word-length metric (Theorem, metric formula): the geodesic length is a minimum over crossing-number and site-pairing data, equal to a relaxed length plus exactly 2 for each isolated cycle that the connectivity (single-Eulerian-trail) constraint forbids. The metric is forced by a strand bound limiting each interface component to at most one up- and one down-crossing per edge, leaving a single unbounded catalytic datum; it reproduces A396406 through u_{22} with no breadth-first search. The kernel depends only on the similarity class of T (Lemma, dilation invariance).

Universality and its sharp threshold. Because the symbolic normal form is triangle-independent and ζ_T is never a root of unity (Niven), two group elements collide under ρ_T only when their symbolic data agree and their translation polynomials agree at $t = \zeta_T$; an extra collision at depth d thus forces a nonzero integer Laurent polynomial of height $\leq 2d$ to vanish at ζ_T. By Gauss's lemma the shape's minimal polynomial $\mu_T(t) = c_T t^2 - e_T t + c_T$ must then divide a realized word-difference, so c_T divides both extreme coefficients and $c_T \leq 2d$. This yields an *effective all-depths universality theorem*: every shape with $c_T > 2d$ satisfies $u_d^T = u_d$ (Theorem, effective universality). The arithmetic of c_T — odd, with all prime factors $\equiv 1 \pmod 4$, and ≥ 5 — leaves only a sparse finite set of candidate shapes at each depth, and a one-sided exact modular certificate over these candidates closes the remaining gap:

> *For every right triangle with positive unequal rational legs, $u_d^T = u_d$ for all $0 \leq d \leq 32$, and the bound is sharp: the $(1,2)$ triangle satisfies $u_{33}^{(1,2)} = u_{33} - 30 < u_{33}$* (Theorem, sharp uniform universality).

Refutation of all-depths universality. The natural conjecture that $u_d^T = u_d$ for *all* d is false for every algebraic shape. Since R_0R_1 is the half-turn about the right-angle vertex, $R_0R_1R_2$ is a glide reflection along the hypotenuse, and its square is the pure translation by $2(t^{-1} - 1)$; conjugating and multiplying these glide-squares realizes the entire ideal $2(t-1)\mathbb{Z}[t^{\pm 1}]$. As that ideal contains a multiple of every μ_T, an explicit word

$$w_T = \tau^{\,c_T}\, r\, \tau^{\,-e_T}\, r\, \tau^{\,c_T}\, r^{-2}, \qquad \tau = (R_0R_1R_2)^2, \quad r = R_x R_h,$$

realizes the translation $2(t^{-1} - 1)\mu_T(t)$, which is nonzero in the generic group but evaluates to 0 at ζ_T, hence lies in $\ker \rho_T$ (Theorem, non-universality). Thus $\ker \rho_T$ is, for every algebraic shape, the principal ideal $2(t-1)\mu_T\,\mathbb{Z}[t^{\pm 1}]$, isomorphic to $\mathbb{Z}[t^{\pm 1}]$ and normally generated by the single witness w_T (Corollary, kernel classification). The first deviation depth is therefore finite and pinned to linear growth in the arithmetic complexity of the shape,

$$\max(33,\, c_T) \;\leq\; n_T \;\leq\; 6(2c_T + |e_T|) + 8 \;\leq\; 24c_T + 8$$

(Corollary, deviation-depth sandwich), with $n_{(1,2)} = 33$ realizing the lower bound; the deviation depth is itself a shortest-vector length in the lamplighter word metric. Consequently the universal sequence is exactly the orbit growth of the generic right triangle, realized precisely by the (uncountably many) transcendental leg ratios, while the isosceles case ($c_T = 1$, deviation at depth 4) is the extreme degeneration to a wallpaper group (Corollary, transcendence dichotomy).

The exact growth rate. The lamp model linearizes the metric into a transfer with one catalytic variable, but the catalytic variable enters through a *dilation* $t \mapsto q^2 t$, so the resulting functional equation for the relaxed generating function is a linear q-difference equation, not a polynomial-kernel equation; the Bousquet-Mélou–Jehanne framework does not apply and the solution is q-holonomic. Unfolding the equation telescopes the relaxed series into explicit rapidly-convergent q-series blocks, and the dominant singularity is the smallest positive root q^* of the travel-block equation $\Sigma_1(q) = 1$. We prove that this rate is also the *true* rate, unconditionally: the travel block is connectivity-invariant (each pure-translation run is a single Euler trail and incurs no penalty), so the true series shares it verbatim; squeezing the true count between its pure-travel subcount and the relaxed series gives

$$\beta_2 \;=\; 1/\sqrt{q^*} \;=\; 1.4916177871143742268\ldots$$

exactly (Proposition, asymptotic ratio), with no dependence on any analytic conjecture. In particular $\beta_2 < 3/2$, excluding the value that the finite-depth ratios — converging at rate $O(1/d)$ with a period-4 oscillation — had suggested.

What is conditional and what is open. The growth rate $\beta_2 = 1.4916177871\ldots$ is determined unconditionally in Section 9 as $1/\sqrt{q^*}$, where q^* is the least positive root of $\Sigma_1(q) = 1$; in particular $\beta_2 \neq 3/2$. Transcendence questions are treated in the companion paper [2], where the growth series of A396406 and of its relaxed companion are proved transcendental over $\mathbb{Q}(x)$ (each catalytic building block unconditionally so, by a Pólya–Carlson natural boundary); this explains the empirical absence, recorded below, of any low-order linear, holonomic, or low-degree algebraic relation over the 43 known terms. The transcendence of the *number* β_2 itself remains open. In higher dimensions — the reflection groups of Euclidean orthoschemes (path simplices), of which the right triangle is the planar case — the abstract right-angled Coxeter envelope has *rational* growth with rate $r_n = 1 + 2\cos\frac{2\pi}{n+3}$, but the geometric group is an amenable proper quotient of it, matching the rational envelope only up to a finite depth before deviating; the plane is the one dimension in which that deviation has been carried past its onset (at depth 10) and found transcendental, and whether higher dimensions are eventually transcendental as well is open [3].

Reproducibility. The eight length-10 affine relations and the shape-independence of u_d through depth 22 are formally verified in the Lean 4 proof assistant; the depth-32 threshold, the depth-38 data provenance of A396406, and the deviation witnesses are certified by exact modular and rational arithmetic, with two independent reproductions. All scripts and Lean sources are described in the reproducibility section.

Related work. The problem sits between Coxeter-group word-growth and the geometry of metabelian groups. Isola and Piergallini [6] show that the reflection group of a generic triangle carries no unexpected relations; the present results are the explicit rational right-triangle analogue, replacing genericity by a Diophantine faithfulness criterion in ζ_T and adding exact word-growth data. The wreath product $\mathbb{Z} \wr \mathbb{Z}$ that emerges is the textbook amenable group of exponential growth; its appearance here, generated by three mirrors of a Euclidean triangle, places the rational shapes inside the program of quantifying group approximations, with the deviation sandwich giving the explicit girth of convergence in the space of marked groups. The geodesic-language and growth analysis is in the spirit of the lamplighter computations of Cleary–Elder–Taback; the catalytic q-difference structure of the growth series, and the consequent exclusion of every elementary closed form for β_2, is the point at which A396406 departs from the rational growth series of the standard lamplighter generating sets. The rotation number ζ_T governing the translation module coincides with the rotation number of Sadun's generalized-pinwheel substitution tilings, though the invariant studied there (tiling-space cohomology) is distinct from the word-growth sequence treated here.

2 Setting and the Coxeter presentation

2.1 The triangle, its mirrors, and the orbit-growth sequence

Let T be a triangle with vertices $V_0, V_1, V_2 \in \mathbb{Q}^2$ and leg lengths $a, b \in \mathbb{Q}_{>0}$, having its right angle at V_2 and *unequal* legs, $a \neq b$. For $i \in \{0, 1, 2\}$ let

$$R_i \in \mathrm{Isom}(\mathbb{R}^2)$$

denote the Euclidean reflection across the line containing the side of T opposite the vertex V_i, and set

$$W(T) \ = \ \langle R_0, R_1, R_2 \rangle \ \leq \ \mathrm{Isom}(\mathbb{R}^2).$$

When explicit coordinates are required we normalize the right-angle vertex to $B = (a, 0)$ with $A = (0, 0)$ and $C = (a, b)$, writing a, b for the two leg lengths; the mirrors R_0, R_1 are then the reflections across the two legs meeting at the right angle.

Remark 1 (Notation dictionary). Throughout the paper the three generators carry two interchangeable names, fixed once here:

$$R_0 = R_x \text{ (leg on the } x\text{-axis)}, \qquad R_1 = R_y \text{ (leg on the } y\text{-axis)}, \qquad R_2 = R_h \text{ (hypotenuse)}.$$

The subscripted names R_0, R_1, R_2 index the opposite vertices V_0, V_1, V_2; the lettered names R_x, R_y, R_h name the mirror lines and are used from §3 onward, where the right-angle vertex is placed at the origin with the two legs along the positive coordinate axes (the normalization $A = (0, 0)$, $B = (a, 0)$, $C = (a, b)$ above, rescaled by $1/a$). The two normalizations differ only by a similarity and induce the same abstract relations and orbit counts.

We study the orbit of T under $W(T)$ through breadth-first search in the Cayley graph of $W(T)$ relative to the generating set $S = \{R_0, R_1, R_2\}$. For $d \geq 0$ put

$$\mathcal{O}_d(T) \ = \ \{\, g \cdot T : g \in W(T), \ \ell_S(g) = d \,\}, \qquad u_d^T \ := \ \big|\mathcal{O}_d(T)\big|,$$

where ℓ_S is word length with respect to S, and $u_0^T = 1$. All distances are computed in exact rational arithmetic, and the relations below are verified symbolically over $\mathbb{Q}(a, b)$. We write $u_d := u_d^T$ when the shape is fixed or immaterial.

Remark 2. Two groups govern u_d. The *abstract* group W is the finitely presented Coxeter group of Theorem 3 below, depending only on the right-angle and non-isosceles hypotheses. The *Euclidean* group $W(T)$ is its image under the affine action $\rho_T \colon W \to \mathrm{Aff}(\mathbb{R}^2)$ induced by the mirrors of T, which depends on the shape. Distinct words of W may act identically on $\mathbb{R}^2$, collapsing orbit elements; the counts u_d are those of the Euclidean orbit (Definition 8).

2.2 The Coxeter presentation

Let Γ be the abstract group obtained from the free product $\langle r_0 \rangle * \langle r_1 \rangle * \langle r_2 \rangle$ of three copies of $\mathbb{Z}/2$ by the canonical map onto $W(T)$ sending $r_i \mapsto R_i$, and let $K \trianglelefteq \langle r_0 \rangle * \langle r_1 \rangle * \langle r_2 \rangle$ be the kernel of this map: K is the set of all relations holding in $\mathrm{Isom}(\mathbb{R}^2)$ among R_0, R_1, R_2. The presentation theorem identifies K.

Theorem 3 (Coxeter presentation)**.** *Let T be a right triangle with the right angle at V_2 and with unequal legs $a \neq b$. Let*

$$W \ := \ \big\langle\, R_0, R_1, R_2 \ \big| \ R_i^2 = 1 \ (i = 0, 1, 2), \ \ (R_0 R_1)^2 = 1 \,\big\rangle$$

be the Coxeter group whose diagram is the path on three nodes with edge labels $m(R_0, R_1) = 2$ and $m(R_0, R_2) = m(R_1, R_2) = \infty$. Then the pairwise rotation orders of the three mirrors of T are exactly $m(R_0, R_1) = 2$, $m(R_0, R_2) = m(R_1, R_2) = \infty$, and consequently $r_i \mapsto R_i$ induces a surjection *$\rho_T : W \twoheadrightarrow W(T)$. This surjection is* not *injective for any admissible shape: the affine action imposes further relations beyond $(R_0R_1)^2 = 1$, so $W(T)$ is a* proper *quotient of W, and* $\ker \rho_T \supsetneq \{1\}$*. The first such relation is the object of Theorem 26; the abstract Coxeter group W therefore serves only as an* envelope *for $W(T)$.*

Proof. Each R_i is a reflection, so $R_i^2 = 1$; the mirrors of R_0, R_1 meet at $\pi/2$, so R_0R_1 is a half-turn and $(R_0R_1)^2 = 1$. Hence all defining relators of W hold in $W(T)$, giving the surjection $\rho_T : W \twoheadrightarrow W(T)$. The remaining interior angles $\arctan(b/a)$, $\arctan(a/b)$ are, by Niven's theorem (a rational multiple of π has rational tangent only for tangent in $\{0, \pm 1\}$, excluded by $a \neq b$), irrational multiples of π; so R_0R_2 and R_1R_2 have infinite order and no bond $(R_iR_2)^m = 1$ is forced. Thus the pairwise orders are exactly as stated, and W is the largest group compatible with them.

The surjection is *not* an isomorphism. Coxeter's structure theorem would identify $W(T)$ with W only if the affine (Tits) representation ρ_T were faithful; but $W(T) \leq \mathrm{Isom}(\mathbb{R}^2)$ is amenable (indeed virtually lamplighter, §8), whereas $W \cong (\mathbb{Z}/2 \times \mathbb{Z}/2) * \mathbb{Z}/2$ contains a free subgroup and is non-amenable. Hence $\ker \rho_T$ is nontrivial and $W(T)$ is a proper quotient. Explicitly, $\ker \rho_T$ is computed in Theorem 26: its shortest element has length $2n_T$, so $W(T)$ and W have the same spherical growth only through depth $n_T - 1$. □

Remark 4. The non-isosceles hypothesis $a \neq b$ is used solely to exclude the angle $\pi/4$ in Theorem 3; for the isosceles right triangle the third relation $(R_0R_2)^4 = 1$ holds and W is a finite Coxeter group, a degenerate case outside the present scope.

2.3 The Steinberg–Poincaré series and the Fibonacci phase

The growth of the abstract layer is governed by the spherical growth series of the Coxeter group W of Theorem 3, i.e. the Poincaré series

$$W(t) \;=\; \sum_{d \geq 0} \big|\{\, g \in W : \ell_S(g) = d \,\}\big|\, t^d.$$

Proposition 5 (Steinberg–Poincaré series)**.** *For the Coxeter group W of Theorem 3,*

$$W(t) \;=\; \frac{(1+t)^2}{1 - t - t^2}.$$

Proof. Steinberg's formula [5] expresses the Poincaré series of a Coxeter system (W, S) as

$$\frac{1}{W(t^{-1})} \;=\; \sum_{\substack{J \subseteq S \\ W_J \text{ finite}}} \frac{(-1)^{|J|}}{W_J(t)},$$

summed over those subsets J generating a finite parabolic W_J, with $W_J(t)$ its (polynomial) Poincaré series. Here the finite parabolics are: $J = \emptyset$ ($W_\emptyset(t) = 1$); the three singletons $\{R_i\}$, each with $W_J(t) = 1 + t$; and the rank-2 subset $J = \{R_0, R_1\}$, which generates the dihedral group of order 4 with $W_J(t) = (1+t)^2$. The subsets $\{R_0, R_2\}$ and $\{R_1, R_2\}$ generate the infinite dihedral group and contribute nothing. Substituting,

$$\frac{1}{W(t^{-1})} = 1 - \frac{3}{1+t} + \frac{1}{(1+t)^2} = \frac{(1+t)^2 - 3(1+t) + 1}{(1+t)^2} = \frac{t^2 - t - 1}{(1+t)^2}.$$

Inverting both sides,

$$W(t^{-1}) \;=\; \frac{(1+t)^2}{t^2-t-1}.$$

Now substitute $t \mapsto t^{-1}$. The numerator becomes $(1+t^{-1})^2 = t^{-2}(1+t)^2$, and the denominator becomes $t^{-2} - t^{-1} - 1 = t^{-2}(1-t-t^2)$; the common factor t^{-2} cancels, giving

$$W(t) \;=\; \frac{(1+t)^2}{1-t-t^2}. \qquad \square$$

Corollary 6 (Abstract layer counts are Fibonacci)**.** *Write F_n for the Fibonacci numbers with the conventions $F_0 = 0$, $F_1 = F_2 = 1$. The coefficients of the Steinberg–Poincaré series are*

$$[t^d]\, W(t) \;=\; F_{d+3} \qquad (d \geq 1),$$

with $[t^0]\, W(t) = 1$.

Proof. Write $(1+t)^2 = 1 + 2t + t^2$ and let $\Phi(t) = 1/(1-t-t^2) = \sum_{n\geq 0} F_{n+1} t^n$, the Fibonacci generating function. Then for $d \geq 1$,

$$[t^d]\, W(t) = F_{d+1} + 2F_d + F_{d-1},$$

where the term F_{d-1} is read with the convention $F_0 = 0$ when $d = 1$. Using $F_{d+1} + F_d = F_{d+2}$ and $F_d + F_{d-1} = F_{d+1}$, this equals $F_{d+2} + F_{d+1} = F_{d+3}$. In particular at $d = 1$ one gets $F_2 + 2F_1 + F_0 = 1 + 2 + 0 = 3 = F_4$, consistent with the direct expansion $W(t) = 1 + 3t + 5t^2 + 8t^3 + \cdots$; the boundary value $[t^0]\, W(t) = 1$ is read off the same expansion. $\square$

The Euclidean orbit counts u_d agree with this baseline until the first Euclidean collision.

Corollary 7 (Fibonacci phase)**.** *Fix the reference triangle $T_0 = (3, 4, 5)$. Then*

$$u_d^{T_0} \;=\; F_{d+3} \qquad (1 \leq d \leq 9),$$

and the first depth at which the affine action collapses an abstract orbit element is $d = 10$, where $u_{10}^{T_0} = 225 < F_{13} = 233$. The same values hold for every right triangle T with unequal rational legs; this shape-independence on the range $1 \leq d \leq 9$ and the depth-10 deficit are established symbolically over $\mathbb{Q}(a, b)$ in §3 (Theorem 10). At $d = 0$ one has $u_0 = 1 \neq F_3 = 2$.

Proof. By Proposition 5 and Corollary 6, the abstract layer count at depth d is F_{d+3}, and $u_d^T \leq F_{d+3}$ for every T since u_d^T counts the image $\rho_T(\mathcal{O}_d)$. For the reference triangle $T_0 = (3, 4, 5)$, exact-rational breadth-first search shows that the map from abstract length-d elements to Euclidean orbit points is injective for $1 \leq d \leq 9$, so $u_d^{T_0} = F_{d+3}$ there, and that $u_{10}^{T_0} = 225 = F_{13} - 8$, exhibiting eight affine collisions at length 10. That these values are independent of the admissible shape is not a consequence of the single search but is proved by the symbolic relation analysis of §3: Theorem 10 shows over $\mathbb{Q}(a, b)$ that no nontrivial affine identification occurs below depth 10 and that exactly eight occur at depth 10 for every admissible T. $\square$

Definition 8 (Fibonacci deficit)**.** The *Fibonacci deficit* of T at depth d is

$$\delta_d^T \;:=\; F_{d+3} - u_d^T \;\geq\; 0,$$

the number of abstract length-d orbit elements identified by the affine action ρ_T. By Corollary 7 and Theorem 10, $\delta_d^T = 0$ for $1 \leq d \leq 9$ and $\delta_{10}^T = 8$ for every admissible T. The Euclidean orbit-growth sequence $(u_d^T)_{d\geq 0}$ is recorded as OEIS `A396406`.

Remark 9. Throughout, $\rho_T\colon W \to \mathrm{Aff}(\mathbb{R}^2)$ denotes the affine action of the abstract Coxeter group $W = \langle R_0, R_1, R_2\rangle$ induced by the mirrors of T, and $\zeta_T = (a+bi)/(a-bi)$ is the rotation number attached to the leg pair (a,b); both enter the analysis from §3 onward. The results of this section — the presentation of Theorem 3, the series of Proposition 5, and the Fibonacci phase of Corollary 7 — are unconditional.

3 The eight length-10 relations and depth-10 completeness

Let $W = \langle\, R_0, R_1, R_2 \mid R_i^2 = 1\ (i = 0,1,2),\ (R_0R_1)^2 = 1 \,\rangle$ be the Coxeter group of §2. Throughout, $R_0 = R_x$ is the reflection across the x-leg, $R_1 = R_y$ the reflection across the y-leg, and $R_2 = R_h$ the reflection across the hypotenuse. For a right triangle T let $\rho_T\colon W \to \mathrm{Aff}(\mathbb{R}^2)$ be its side-reflection action. Fix the placement of §2,

$$A = (0,0), \qquad B = (a,0), \qquad C = (a,b), \qquad a,b > 0,$$

so that each generator R_i acts as an affine isometry whose 3×3 matrix has entries in $\mathbb{Q}(a,b)$, the denominators dividing a power of a^2+b^2. For a word $w = w[0]\,w[1]\cdots w[\ell-1]$ in the R_i we write $\rho_T(w) = R_{w[0]}\,R_{w[1]}\cdots R_{w[\ell-1]}$ for the associated affine matrix.

The Poincaré series of the abstract Coxeter group W gives F_{d+3} canonical words of length d for $d \geq 1$; in particular there are $F_{13} = 233$ canonical words of length 10. Exact-rational breadth-first enumeration of the orbit of a non-isosceles right triangle gives $u_{10} = 225$, a deficit of 8. The eight collapsing pairs are the length-10 relations of Table 1; they are the first relations of the side-reflection action beyond the abstract Coxeter relations.

#	word A	word B
1	$R_0R_1R_2R_0R_2R_0R_1R_2R_1R_2$	$R_2R_1R_2R_0R_1R_2R_0R_2R_0R_1$
2	$R_0R_1R_2R_1R_2R_0R_1R_2R_0R_2$	$R_2R_0R_2R_0R_1R_2R_1R_2R_0R_1$
3	$R_0R_2R_0R_1R_2R_1R_2R_0R_1R_2$	$R_2R_0R_1R_2R_1R_2R_0R_1R_2R_0$
4	$R_0R_2R_0R_2R_0R_1R_2R_1R_2R_0$	$R_1R_2R_1R_2R_0R_1R_2R_0R_2R_1$
5	$R_0R_2R_0R_2R_0R_1R_2R_1R_2R_1$	$R_1R_2R_1R_2R_0R_1R_2R_0R_2R_0$
6	$R_0R_2R_1R_2R_0R_1R_2R_0R_2R_0$	$R_1R_2R_0R_2R_0R_1R_2R_1R_2R_1$
7	$R_0R_2R_1R_2R_0R_1R_2R_0R_2R_1$	$R_1R_2R_0R_2R_0R_1R_2R_1R_2R_0$
8	$R_1R_2R_0R_1R_2R_0R_2R_0R_1R_2$	$R_2R_0R_1R_2R_0R_2R_0R_1R_2R_1$

Table 1: The eight length-10 affine relations. In each row, the two words have the same affine matrix in the side-reflection action ρ_T of W on every right triangle T. The generators are $R_0 = R_x$, $R_1 = R_y$, $R_2 = R_h$ (reflections across the x-leg, y-leg, and hypotenuse).

Theorem 10 (Eight length-10 affine relations)**.** *Each of the eight identities of Table 1 holds for every right triangle. For the placement $A = (0,0)$, $B = (a,0)$, $C = (a,b)$ with $a,b > 0$, the affine matrices*

$$M_A = R_{w_A[0]}\cdots R_{w_A[9]}, \qquad M_B = R_{w_B[0]}\cdots R_{w_B[9]}$$

of the two words w_A, w_B in each row satisfy $M_A = M_B$ as 3×3 affine matrices over $\mathbb{Q}(a,b)$.

Proof. The statement is a finite algebraic identity. Each reflection R_i acts as an affine isometry whose matrix entries are rational functions of a,b over $\mathbb{Q}$, with denominators dividing a power of a^2+b^2. Hence each entry of $M_A - M_B$ is a rational function in $\mathbb{Q}(a,b)$; clearing denominators yields, for each row, a polynomial identity in $\mathbb{Q}[a,b]$. These identities are verified deterministically in exact arithmetic over $\mathbb{Q}(a,b)$.

The eight identities are moreover *formally verified* in the Lean 4 proof assistant. The theorems

`rel1_symbolic`, ..., `rel8_symbolic` in `lean/with_mathlib/SymbolicVerification.lean`

each discharge the six affine-matrix-entry identities over $\mathbb{Q}[a,b]$ that together express $M_A = M_B$. Each such identity is proved by the `ring` tactic alone. The `ring` tactic produces a proof term that is checked by the Lean kernel; the trust base is therefore the Lean kernel together with the `Mathlib` definitions of $\mathbb{Q}[a,b]$ and its commutative-ring structure, with no appeal to numerical sampling, decision heuristics, or specialization to particular legs. Consequently each identity holds as an equality of affine matrices over $\mathbb{Q}(a,b)$, hence for every right triangle. □ □

Remark 11. Because Theorem 10 holds over $\mathbb{Q}(a,b)$, the eight collisions lie in $\ker \rho_T$ for every T; the enumerative deficit $233 - 225 = 8$ is therefore shape-independent.

Proposition 12 (Completeness at depth 10)**.** *Let T be an unequal-leg ($a \neq b$) right triangle. Among the* 233 *canonical Coxeter words of length* 10 *in W, exactly* 225 *have pairwise distinct images under ρ_T: there are* 217 *singleton classes and* 8 *collision pairs, each pair being one row of Table 1. No length-*10 *affine identity exists beyond these eight relations and the abstract Coxeter relations. (For the isosceles triangle $a = b$ additional collisions occur, by Remark 4; the count* 225 *and the distinctness of the* 217 *singleton classes are asserted only for $a \neq b$.)*

Proof. Completeness follows from two ingredients: (a) the computed block partition on the reference triangle $T_0 = (3,4,5)$, and (b) Theorem 10.

All 233 canonical Coxeter words of length 10 are enumerated. For each, the affine matrix $\rho_{T_0}(w)$ is computed in exact rational arithmetic on T_0, and the words are grouped by matrix. The resulting partition has exactly 225 blocks: eight blocks of size 2 and 217 blocks of size 1. The eight size-2 blocks are the rows of Table 1. By Theorem 10 these eight identities hold symbolically over $\mathbb{Q}(a,b)$, so the eight pairs are genuine collisions for every right triangle. Conversely, any further length-10 collision would produce a coincidence of affine matrices already on T_0 in exact arithmetic, contradicting the computed block sizes. Since T_0 is unequal-leg, the 217 singleton blocks remain singletons under specialization to any unequal-leg T. Hence the eight relations, together with the abstract Coxeter relations used to form the canonical words, account for all length-10 affine identities.

As corroboration, each size-2 block was independently re-checked on six further triangles,

$$(5,12), \quad (8,15), \quad (7,24), \quad (1,2), \quad (2,3), \quad (1,7),$$

and in all cases the two words of the block produce the same affine matrix; this is redundant once Theorem 10 is invoked. □ □

4 The triangle-independent symbolic normal form

The side-reflection action carries a triangle-independent symbolic normal form (Lemma 13) in which the shape of the triangle enters only through the evaluation of integer Laurent polynomials at the rotation number ζ_T. Collisions between group elements reduce to the vanishing of bounded-height Laurent polynomials at ζ_T (Proposition 16), a divisibility condition in $\mathbb{Z}[i]$ that drives the effective universality theorem of §5.

4.1 Complex coordinates and the generators

Throughout, the three side-reflections are written interchangeably as $(R_0, R_1, R_2) = (R_x, R_y, R_h)$, where $R_0 = R_x$ is the reflection in the leg on the positive real axis, $R_1 = R_y$ the reflection in the leg on the positive imaginary axis, and $R_2 = R_h$ the reflection in the hypotenuse; this matches the labelling R_0, R_1, R_2 of §5.

Identify $\mathbb{R}^2 = \mathbb{C}$, place the right angle at the origin with the legs along the positive coordinate axes, and rescale by $1/a$, where a is the length of the leg on the positive real axis. This is the normalization used in §5; it agrees, after the affine change placing the right-angle vertex at the origin, with the normalization $A = (0,0)$, $B = (a,0)$, $C = (a,b)$ of the introductory sections. Neither the choice of origin and axes nor the rescaling changes word-lengths or the collision structure of the orbit, so no generality is lost. Let $\sigma(z) = \bar{z}$ denote complex conjugation, and set

$$\zeta = \zeta_T = \frac{a+bi}{a-bi},$$

the rotation number of the triangle with legs (a,b). Since numerator and denominator are complex conjugates, $|\zeta_T| = 1$; we use this standing fact below. In these coordinates the three side-reflections are

$$R_x : z \mapsto \sigma(z), \qquad R_y : z \mapsto -\sigma(z), \qquad R_h : z \mapsto \zeta^{-1}\sigma(z) + \left(1 - \zeta^{-1}\right),$$

where R_x and R_y are the reflections in the two leg lines and R_h is the reflection in the hypotenuse line. We write W for the group they generate and $\rho_T \colon W \to \mathrm{Isom}(\mathbb{C})$ for the resulting faithful action on the plane of the triangle T.

Every element of $\mathrm{Isom}(\mathbb{C})$ produced by composing these generators is an orientation-reversing-or-preserving affine map of the form $z \mapsto \alpha\,\sigma^{\delta}(z) + \beta$ with $\delta \in \{0,1\}$. The normal form below records that the linear part α is always of the shape $\varepsilon\,\zeta^k$ with $\varepsilon \in \{\pm 1\}$ and $k \in \mathbb{Z}$, and that the translation part β is the value at ζ of an integer Laurent polynomial whose support and height are controlled by the word-length.

4.2 The normal form

Lemma 13 (Symbolic normal form)**.** *For every $w \in W$ of word-length ℓ there exist $\varepsilon \in \{\pm 1\}$, $\delta \in \{0,1\}$, an integer $k \in \mathbb{Z}$ with $|k| \le \ell$, and an integer Laurent polynomial $P_w \in \mathbb{Z}[t^{\pm 1}]$ — all four* independent of the triangle T *— such that*

$$\rho_T(w) \colon z \;\longmapsto\; \varepsilon\,\zeta^k\,\sigma^{\delta}(z) \;+\; P_w(\zeta). \tag{1}$$

Moreover

$$\operatorname{supp} P_w \subseteq [-\ell, \ell], \qquad \|P_w\|_\infty \le \ell, \qquad P_w(1) = 0,$$

where $\|P_w\|_\infty$ denotes the maximum absolute value of the coefficients of P_w.

We refer to the triple (ε, δ, k) as the *symbolic class* of w and to P_w as its *translation polynomial.* The defining identity (1) is purely formal: the symbol ζ is treated as an indeterminate t, conjugation σ acts on Laurent polynomials by the substitution $t \mapsto t^{-1}$, and T-dependence is confined to the final evaluation $t = \zeta_T$. The identity $\sigma\big(P(\zeta)\big) = P(\zeta^{-1})$ used in the proof is the analytic content of this convention, valid because $|\zeta_T| = 1$ and the coefficients of P are real, so that $\overline{\zeta} = \zeta^{-1}$.

Proof. The argument is by induction on the word-length ℓ, using the composition law for the affine maps in (1).

Base case. For the three generators the symbolic data are

$$R_x\colon (\varepsilon,\delta,k,P) = (1,\,1,\,0,\,0), \quad R_y\colon (-1,\,1,\,0,\,0), \quad R_h\colon (1,\,1,\,-1,\,1-t^{-1}),$$

read off directly from the formulas for R_x, R_y, R_h. Each has $|k| \le 1$, support contained in $[-1,1]$, height at most 1, and a translation polynomial vanishing at $t=1$ (the constant 0 and $1-t^{-1}$ both do).

Inductive step. Let g be a generator with data $(\varepsilon_g, \delta_g, k_g, P_g)$ and let w have data $(\varepsilon_w, \delta_w, k_w, P_w)$ satisfying the asserted bounds for its length ℓ. Composing the two affine maps

$$\rho_T(g)\colon z \mapsto \varepsilon_g\,\zeta^{k_g}\sigma^{\delta_g}(z) + P_g(\zeta), \qquad \rho_T(w)\colon z \mapsto \varepsilon_w\,\zeta^{k_w}\sigma^{\delta_w}(z) + P_w(\zeta),$$

and using $\sigma(\zeta) = \bar{\zeta} = \zeta^{-1}$ (valid since $|\zeta_T| = 1$) together with the reality of the coefficients of P_w (so that $\sigma\big(P_w(\zeta)\big) = P_w(\zeta^{-1})$), the product gw acts by

$$z \;\longmapsto\; \varepsilon_g\varepsilon_w\,\zeta^{\,k_g \pm k_w}\,\sigma^{\delta_g+\delta_w}(z) \;+\; \Big(P_g(\zeta) + \varepsilon_g\,\zeta^{k_g}\,P_w(\zeta^{\pm1})\Big),$$

where the exponents $\delta_g + \delta_w$ and $\sigma^{\delta_g+\delta_w}$ are read modulo 2, and the sign in $k_g \pm k_w$ and in the argument $\zeta^{\pm1}$ of P_w is -1 exactly when $\delta_g = 1$ — the case in which the conjugation in $\rho_T(g)$ inverts ζ. Hence the symbolic data of gw are

$$\varepsilon_{gw} = \varepsilon_g\varepsilon_w \in \{\pm1\}, \qquad \delta_{gw} = \delta_g + \delta_w \pmod 2 \in \{0,1\}, \qquad k_{gw} = k_g \pm k_w,$$

and the translation polynomial obeys the composition law

$$P_{gw}(t) \;=\; P_g(t) \;+\; \varepsilon_g\,t^{k_g}\,P_w(t^{\pm1}), \tag{2}$$

all of which are manifestly independent of T.

It remains to verify the three bounds for the word gw of length $\ell+1$. Since $|k_g| \le 1$, the exponent satisfies $|k_{gw}| = |k_g \pm k_w| \le 1 + \ell = \ell + 1$. For the translation polynomial, the operations applied to P_w in (2) are: substitution $t \mapsto t^{\pm1}$, multiplication by the monomial $\pm t^{k_g}$, and addition of the generator's polynomial P_g. Inverting t permutes the support symmetrically and leaves the multiset of coefficients — hence the height — unchanged; multiplying by $\pm t^{k_g}$ with $|k_g| \le 1$ shifts the support by at most 1 and again preserves the height; and adding P_g, which has height at most 1 and support in $[-1,1]$, raises the height by at most 1 and enlarges the support radius by at most 1. Therefore

$$\|P_{gw}\|_\infty \le \|P_w\|_\infty + 1 \le \ell + 1, \qquad \operatorname{supp} P_{gw} \subseteq [-(\ell+1),\,\ell+1].$$

Finally, evaluating (2) at $t=1$ gives $P_{gw}(1) = P_g(1) + \varepsilon_g\,P_w(1) = 0 + 0 = 0$, since both $P_g(1) = 0$ (base case) and $P_w(1) = 0$ (induction hypothesis). This completes the induction. □

Remark 14. The composition law (2) controls the growth of P_w: appending a generator either fixes the support of the translation polynomial or shifts it by one and adds a height-one correction, so the support radius and the height each grow by at most one per letter. The vanishing $P_w(1) = 0$ is the algebraic shadow of the geometric fact that the degenerate limit $\zeta \to 1$ (an isoceles right triangle, where $a = b$) collapses the translation part: every word fixes the affine structure of the degenerate triangle, consistently with the exclusion $a \ne b$ throughout.

4.3 Reduction of collisions to polynomial vanishing

The normal form converts the geometric question of when two group elements have equal action into an arithmetic question about a single Laurent polynomial. The key separation is provided by the irrationality of the rotation angle.

Lemma 15 (Class separation)**.** *The rotation number ζ_T is not a root of unity. Consequently the linear parts $\varepsilon\,\zeta_T^k$ arising in (1), ranging over $\varepsilon \in \{\pm 1\}$ and $k \in \mathbb{Z}$, are pairwise distinct; and two elements $w, w' \in W$ satisfy $\rho_T(w) = \rho_T(w')$ if and only if they share the same symbolic class (ε, δ, k) and their translation polynomials agree at $t = \zeta_T$:*

$$\rho_T(w) = \rho_T(w') \iff (\varepsilon, \delta, k)_w = (\varepsilon, \delta, k)_{w'} \quad \text{and} \quad P_w(\zeta_T) = P_{w'}(\zeta_T). \tag{3}$$

Proof. For legs with $a \neq b$ the angle $\arg \zeta_T = 2\arctan(b/a)$ is an irrational multiple of π by Niven's theorem on rational multiples of π with rational tangent, so ζ_T is not a root of unity and $|\zeta_T| = 1$ with $\zeta_T \neq \pm 1$. Hence the powers ζ_T^k, $k \in \mathbb{Z}$, are pairwise distinct, and none equals $-\zeta_T^{k'}$ for any k': from $-\zeta_T^k = \zeta_T^{k'}$ we would get $\zeta_T^{2(k-k')} = 1$, making ζ_T a root of unity, a contradiction. Two affine maps $z \mapsto \alpha\,\sigma^\delta(z) + \beta$ coincide if and only if their linear parts $\alpha\,\sigma^\delta$ and their translation parts β coincide. By the previous sentence, equality of the linear parts forces $\delta_w = \delta_{w'}$, $k_w = k_{w'}$, and $\varepsilon_w = \varepsilon_{w'}$; equality of the translation parts is exactly $P_w(\zeta_T) = P_{w'}(\zeta_T)$. □

Let u_d^{sym} denote the number of distinct symbolic tuples $(\varepsilon, \delta, k, P_w)$ realized by words of length at most d. This count is a T-independent integer, and it equals the generic depth-d orbit count appearing in the effective universality theorem (Theorem 19): distinct symbolic data give distinct generic isometries, while any extra coincidence for a particular triangle can only *decrease* the count below u_d^{sym}. Combining this with Lemma 15 yields the precise mechanism by which a triangle can fail universality.

Proposition 16 (Collisions as polynomial vanishing)**.** *Fix a triangle T with rotation number ζ_T, and let u_e^T denote the depth-e orbit count for T. If $u_e^T \neq u_e^{\mathrm{sym}}$ for some $e \leq d$, then there exist two words w, w' of length at most e lying in the same symbolic class such that the nonzero integer Laurent polynomial*

$$D \;=\; P_w - P_{w'} \;\in\; \mathbb{Z}[t^{\pm 1}], \qquad D \not\equiv 0,$$

satisfies

$$D(\zeta_T) = 0, \qquad \operatorname{supp} D \subseteq [-e,\, e], \qquad \|D\|_\infty \leq 2e, \qquad D(1) = 0. \tag{4}$$

Proof. A deficiency $u_e^T < u_e^{\mathrm{sym}}$ means that two words $w \neq w'$ with distinct symbolic data nevertheless have $\rho_T(w) = \rho_T(w')$. By the collision criterion (3) they must then share the same symbolic class (ε, δ, k) — so their distinctness lies in the translation polynomials — and satisfy $P_w(\zeta_T) = P_{w'}(\zeta_T)$. Set $D = P_w - P_{w'}$. Then $D \not\equiv 0$ as Laurent polynomials (else the symbolic data would coincide), $D(\zeta_T) = 0$, and the support and height bounds follow from Lemma 13: each of $P_w, P_{w'}$ has support in $[-e, e]$ and height at most e, so D has support in $[-e, e]$ and height at most $2e$. Finally $D(1) = P_w(1) - P_{w'}(1) = 0$. □

Proposition 16 is the bridge from geometry to arithmetic on which §5 rests: a failure of universality at depth e for the triangle T is *equivalent* to the existence of a nonzero integer Laurent polynomial of support radius at most e, height at most $2e$, and vanishing at $t = 1$, that also vanishes at the rotation number ζ_T. Because the height and support of D are bounded explicitly in terms of the depth alone — with no dependence on T — this condition can be tested against the algebraic constraints imposed by ζ_T. Since ζ_T is an algebraic number of degree 2 over $\mathbb{Q}$, vanishing $D(\zeta_T) = 0$

forces the minimal polynomial $\mu_T \in \mathbb{Z}[t]$ of ζ_T to divide a suitable integer polynomial associated with D; by Gauss's lemma this divisibility takes place in $\mathbb{Z}[t]$, so the leading coefficient c_T of μ_T divides the extreme coefficients of D, whence $c_T \le \|D\|_\infty \le 2e$. No triangle whose rotation number has $c_T > 2e$ can therefore admit a collision at depth e, which is the divisibility obstruction exploited in §5.

5 Effective universality and the sharp depth-32 threshold

The deviation depth of an arbitrary unequal-leg shape is controlled by an arithmetic mechanism: at each fixed depth, universality reduces to a finite, explicitly sparse list of arithmetically small shapes (Theorem 19). The orbit count is shape-independent for $0 \le d \le 32$ and first fails at depth 33 for the $(1,2)$ triangle (Theorem 20): there $u_{33}^{(1,2)} = 3848354$ against the universal value $u_{33} = 3848384$, a deficit of exactly 30.

Throughout, $W = \langle R_0, R_1, R_2 \mid R_i^2 = 1,\ (R_0R_1)^2 = 1\rangle$. The three involutions are identified with the side-reflections of the right triangle by the fixed dictionary

$$R_0 = R_x \text{ (leg on the } x\text{-axis)}, \qquad R_1 = R_y \text{ (leg on the } y\text{-axis)}, \qquad R_2 = R_h \text{ (hypotenuse)},$$

with the right angle placed at the origin and the legs along the positive coordinate axes, rescaled so that the x-leg has unit length. For a right-triangle parameter $T = (a, b)$ with positive rational legs $a \ne b$ we write ρ_T for the side-reflection realization and $u_d^T := |\{\rho_T(w) : |w|_W = d\}|$ for its orbit-growth sequence. We write u_d for the generic (shape-independent) value, namely the common value of u_d^T wherever it is established to be T-independent.

5.1 The arithmetic complexity of a shape

Identify $\mathbb{R}^2 = \mathbb{C}$, place the right angle at the origin with the legs along the positive coordinate axes, and rescale by $1/a$; neither operation changes word-lengths or collision structure. Writing $\sigma(z) = \bar{z}$ and $\zeta = \zeta_T = (a + bi)/(a - bi)$ for the *rotation number*, the three side-reflections are

$$R_x : z \mapsto \sigma(z), \qquad R_y : z \mapsto -\sigma(z), \qquad R_h : z \mapsto \zeta^{-1}\sigma(z) + \bigl(1 - \zeta^{-1}\bigr),$$

the reflections in the two leg lines and in the hypotenuse line, respectively. By the symbolic normal form (Lemma 13), every $w \in W$ of word-length ℓ acts by

$$\rho_T(w)\colon z \;\longmapsto\; \varepsilon\, \zeta^k\, \sigma^\delta(z) \;+\; P_w(\zeta), \qquad \varepsilon \in \{\pm 1\},\ \delta \in \{0, 1\},\ k \in \mathbb{Z},\ |k| \le \ell,$$

with $P_w \in \mathbb{Z}[t^{\pm 1}]$ *independent of* T, $\operatorname{supp} P_w \subseteq [-\ell, \ell]$, every coefficient of P_w bounded in absolute value by ℓ, and $P_w(1) = 0$. The shape T enters only through the evaluation of these triangle-independent Laurent polynomials at the single point ζ_T.

Definition 17 (Arithmetic complexity and minimal polynomial)**.** Scale the legs to coprime integers (a, b) with $a \ne b$, and write $\zeta_T = \mu/\lambda$ in lowest terms in $\mathbb{Z}[i]$, so that

$$\lambda = a - bi \quad (a + b \text{ odd}), \qquad \lambda = \frac{a - bi}{1 + i} \quad (a, b \text{ both odd}).$$

The *arithmetic complexity* of T is $c_T := N(\lambda)$, the norm of λ in $\mathbb{Z}[i]$. The *minimal polynomial* of the shape is the primitive integer polynomial

$$\mu_T(t) \;=\; c_T\, t^2 - e_T\, t + c_T \;\in\; \mathbb{Z}[t],$$

with roots $\zeta_T^{\pm 1}$, where $e_T = 2(a^2 - b^2)$ when $a + b$ is odd and $e_T = a^2 - b^2$ when a, b are both odd.

For example, $(a, b) = (1, 2)$ gives $\zeta_T = (-3 + 4i)/5$, $c_T = 5$, and $\mu_T = 5t^2 + 6t + 5$, while $(a, b) = (3, 4)$ gives $\zeta_T = (-7 + 24i)/25$, $c_T = 25$, and $\mu_T = 25t^2 + 14t + 25$.

Lemma 18 (Arithmetic of the complexity)**.** *With (a, b) coprime and $a \neq b$:*

(i) *c_T is odd;*

(ii) *every prime factor of c_T is $\equiv 1 \pmod 4$;*

(iii) *$c_T \geq 5$;*

(iv) *μ_T is primitive.*

For general coprime (a, b) (not assuming $a \neq b$), $c_T = 1$ if and only if $a = b$, in which case $\zeta_T \in \{\pm i\}$ is a root of unity.

Proof. (i) If $a + b$ is odd then $c_T = a^2 + b^2$ is odd; if a, b are both odd then $a^2 + b^2 \equiv 2 \pmod 8$, so $c_T = (a^2 + b^2)/2$ is odd.

(ii) A rational (inert) prime $q \equiv 3 \pmod 4$ dividing $c_T = \lambda\bar{\lambda}$ would divide λ itself; but then q would divide both the real and imaginary parts of λ, hence both a and b, contradicting $\gcd(a, b) = 1$. Thus every odd prime factor of c_T splits in $\mathbb{Z}[i]$, i.e. is $\equiv 1 \pmod 4$.

(iii) By (i) and (ii), c_T is an odd integer all of whose prime factors are $\equiv 1 \pmod 4$; the smallest such integer exceeding 1 is 5. For general coprime (a, b), $c_T = 1$ iff ζ_T is a unit of $\mathbb{Z}[i]$, i.e. $\zeta_T \in \{\pm 1, \pm i\}$; for positive legs this forces $a = b$. Under the hypothesis $a \neq b$ this case is excluded, so $c_T \geq 5$.

(iv) An odd prime dividing both c_T and e_T would divide both $a^2 + b^2$ and $a^2 - b^2$, hence both a and b, again contradicting coprimality. $\square$

5.2 Effective universality

Since ζ_T is non-real with $a, b > 0$, its powers $\pm\zeta_T^k$ are pairwise distinct (Niven's theorem rules out roots of unity), so two group elements can have equal ρ_T-images only if their symbolic data agree in (ε, δ, k) and their translation polynomials agree at $t = \zeta_T$. Writing u_d^{sym} for the number of distinct symbolic tuples at depth d — a T-independent integer equal to the generic count u_d — any deviation $u_e^T \neq u_e$ at some depth $e \leq d$ forces a *nonzero* difference

$$D \;=\; P_w - P_{w'} \;\in\; \mathbb{Z}[t^{\pm 1}], \qquad D \neq 0, \qquad D(\zeta_T) = 0, \tag{5}$$

of two same-symbolic-class ball elements of depth $\leq d$.

Theorem 19 (Effective universality)**.** *Let $T = (a, b)$ have coprime positive legs with $a \neq b$, and let c_T, μ_T be as in Definition 17. If $u_e^T \neq u_e$ for some $e \leq d$, then μ_T divides a realized word-difference in $\mathbb{Z}[t]$; in particular c_T divides* both extreme coefficients *of that difference, so $c_T \leq 2d$. Consequently every triangle with $c_T > 2d$ satisfies universality at all depths $\leq d$; equivalently, for every unequal-leg rational triangle,*

$$u_d^T = u_d \qquad \textit{for all } d < \tfrac{1}{2}\, c_T\,.$$

Proof. Take $D \neq 0$ as in (5), built from two ball elements of depth $\leq d$; by Lemma 13 every coefficient of D is bounded by $2d$ in absolute value. Normalize $\widetilde{D} = t^m D \in \mathbb{Z}[t]$ with $\widetilde{D}(0) \neq 0$; this multiplication by a unit of $\mathbb{Z}[t^{\pm 1}]$ changes neither the set of extreme coefficients nor the value at ζ_T. Since ζ_T is non-real, its minimal polynomial over $\mathbb{Q}$ is the degree-2 polynomial μ_T, with conjugate root $\overline{\zeta_T} = \zeta_T^{-1}$. As $\widetilde{D}$ has integer coefficients and vanishes at ζ_T, Gauss's lemma gives a factorization $\widetilde{D} = \mu_T \cdot E$ with $E \in \mathbb{Z}[t]$. Comparing leading and constant coefficients,

$$\mathrm{lead}(\widetilde{D}) = c_T\ \mathrm{lead}(E), \qquad \mathrm{const}(\widetilde{D}) = c_T\ \mathrm{const}(E),$$

so c_T divides both extreme coefficients of $\widetilde{D}$ and

$$c_T \ \le\ |\,\mathrm{lead}(\widetilde{D})| \ \le\ 2d.$$

The contrapositive is the stated universality bound. □

Theorem 19 is an *all-depths* universality statement: at each fixed depth d it reduces the question to the finitely many shapes with $c_T \le 2d$, and by Lemma 18 these are very sparse. The admissible complexities are the odd integers all of whose prime factors are $\equiv 1 \pmod 4$; the smallest are

$$5,\ 13,\ 17,\ 25,\ 29,\ 37,\ 41,\ 53,\ 61,\ 65,\ \ldots,$$

so that $45 = 3^2 \cdot 5$ and $49 = 7^2$ are excluded because 3 and 7 are inert in $\mathbb{Z}[i]$. The cutoff $c_T \le 2d$ pins the candidate list at each depth:

d	$2d$	admissible $c_T \le 2d$
30	60	$\{5, 13, 17, 25, 29, 37, 41, 53\}$
32	64	$\{5, 13, 17, 25, 29, 37, 41, 53, 61\}$
36	72	$\{5, 13, 17, 25, 29, 37, 41, 53, 61, 65\}$

Each admissible complexity is realized by finitely many rotation numbers ζ_T (sixteen for $d = 30$; twenty-four for $d = 36$), and these finitely many candidate shapes can be checked mechanically.

5.3 The finite arithmetic certificate

For each candidate rotation number ζ with $c_T \le 2d$, evaluate the entire symbolic ball at ζ in a finite field $\mathbb{F}_q$, chosen with $q \equiv 1 \pmod 4$ so that $i \in \mathbb{F}_q$, and with $q \nmid c_T$. This evaluation is *one-sided exact*: two symbolic images that are distinct modulo q are necessarily distinct in $\mathbb{Q}(i)$. Thus a collision-free ball mod q certifies a collision-free ball over $\mathbb{Q}(i)$, with no false negatives. Modular arithmetic is therefore a fast *filter*: a candidate that is collision-free under the modular evaluation is settled, while a candidate exhibiting a modular collision is referred to exact arithmetic over $\mathbb{Q}(i)$, which decides it without error. The settled threshold rests on exact $\mathbb{Q}(i)$ confirmation of the finite candidate list; the modular filter, run over several independent primes, only accelerates the search.

At depth 30 the certificate evaluates a ball of 3,336,511 symbolic elements at the 16 candidate rotation numbers with $c_T \le 60$ over independent 31-bit primes; every candidate is collision-free, and the surviving list is confirmed by exact arithmetic over $\mathbb{Q}(i)$. Combined with Theorem 19 for the shapes with $c_T > 60$, this controls all depths ≤ 30; the depth-32 threshold is obtained from the larger run described below.

Theorem 20 (Sharp uniform universality at depth 32)**.** *For every right triangle with positive unequal rational legs,*

$$u_d^T = u_d \qquad \textit{for all } 0 \le d \le 32.$$

The bound is sharp*: the* $(1,2)$ *triangle satisfies*

$$u_{33}^{(1,2)} \ =\ u_{33} - 30 \ =\ 3{,}848{,}354 \ <\ u_{33},$$

so depth 32 *is the exact threshold below which the orbit count is shape-independent.*

Proof. Fix $d = 32$. By Theorem 19, any shape with $u_e^T \neq u_e$ for some $e \leq 32$ has $c_T \leq 2d = 64$; by Lemma 18 the admissible complexities with $c_T \leq 64$ are $\{5, 13, 17, 25, 29, 37, 41, 53, 61\}$, realized by an explicit finite list of rotation numbers. The arithmetic certificate evaluates the symbolic ball through depth 32 at each such ζ over independent 31-bit primes as a fast filter, and the surviving candidate list is confirmed by exact arithmetic over $\mathbb{Q}(i)$; no collision occurs among the admissible shapes. The certified run covers all rotation numbers with $c_T \leq 72 \geq 64$ through depth 32 (Trust base (b)), so the cutoff $c_T \leq 64$ required here lies strictly within the certified range. Hence no admissible shape deviates at any depth ≤ 32, and $u_d^T = u_d$ for $0 \leq d \leq 32$. The deviation at depth 33 for $(1, 2)$ ($c_T = 5$) is realized explicitly by the construction of Theorem 26, giving $u_{33}^{(1,2)} = u_{33} - 30 = 3{,}848{,}354$ and showing the threshold cannot be raised. □

5.4 The translation lattice

The kernel classification, the lamplighter structure, and the membership criterion of the downstream sections all rest on an exact description of the translation lattice

$$\mathcal{T} \;:=\; \{\, P_w(t) \in \mathbb{Z}[t^{\pm 1}] : w \in \ker \text{ on the linear part, } \rho_T(w) \text{ a pure translation} \,\}.$$

Theorem 21 (Translation-lattice equality)**.** $\mathcal{T} = 2(t-1)\,\mathbb{Z}[t^{\pm 1}]$.

Proof. ($\supseteq$) The inclusion $2(t-1)\,\mathbb{Z}[t^{\pm 1}] \subseteq \mathcal{T}$ is the glide-square computation of Lemma 24: the commutator/glide relations of W produce the pure translation by $2(t-1)$, and the $\mathbb{Z}[t^{\pm 1}]$-module action by ζ-multiplication (conjugation by $R_h R_x$) generates the full ideal multiple.

($\subseteq$) Conversely, let $P_w \in \mathcal{T}$, so $\rho_T(w) : z \mapsto z + P_w(\zeta)$ is a pure translation for the shape, and $P_w \in \mathbb{Z}[t^{\pm 1}]$ with $P_w(1) = 0$ by the normal form (Lemma 13). The condition $P_w(1) = 0$ gives $(t-1) \mid P_w$ in $\mathbb{Q}[t^{\pm 1}]$, and since P_w has integer coefficients and $t-1$ is primitive, Gauss's lemma yields $P_w = (t-1)\,Q$ with $Q \in \mathbb{Z}[t^{\pm 1}]$. The parity constraint, that every realized translation polynomial reduces to 0 modulo 2 after dividing by $t-1$ — a consequence of the $(R_i R_j)^2$ relations forcing even total exponent count in each generator (Lemma 13) — gives $Q \in 2\,\mathbb{Z}[t^{\pm 1}]$. Hence $P_w \in 2(t-1)\,\mathbb{Z}[t^{\pm 1}]$, completing the reverse inclusion. □

5.5 Trust bases

The certified content of this section rests on two distinct, independently auditable trust bases, which we keep separate.

(a) **Lean kernel (depths ≤ 22).** The T-independence of the orbit-growth sequence is machine-verified for $0 \leq d \leq 22$ in Lean 4 (theorem `universal_layers_through_22` in `lean/with_mathlib/Computabl`
The proof performs a single breadth-first sweep over the canonical Coxeter words of W, deduplicating the symbolic affine images $\rho_{(a,b)}(w) \in \mathrm{Aff}(\mathbb{Q}(a,b))$ by their numerators relative to the common denominator $(a^2+b^2)^{22}$, and certifies the layer counts

$$\begin{gathered}1,\ 3,\ 5,\ 8,\ 13,\ 21,\ 34,\ 55,\ 89,\ 144,\ 225,\ 351,\\ 554,\ 875,\ 1345,\ 2066,\ 3203,\ 4971,\ 7574,\ 11543,\ 17683,\ 27108,\ 41067\end{gathered}$$

simultaneously for every right triangle with positive unequal legs. This is a kernel-checked statement over $\mathbb{Q}(a,b)$, with no reliance on any single specialization, and it establishes Theorem 19 through depth 22 via the length-10 relations of Theorem 10; it does *not* establish the all-depths statement.

(b) **Modular filter with exact confirmation (depths ≤ 36).** Beyond the Lean kernel, the finite candidate list is settled by the arithmetic certificate, whose guarantee is one-sided exactness of the finite-field evaluation backed by exact arithmetic over $\mathbb{Q}(i)$ on the surviving candidates. A Python/NumPy implementation (`code/zeta_probe/certify.py`) realizes the certificate at depth 30: it evaluates the symbolic ball at the 16 candidate rotation numbers with $c_T \leq 60$ over independent 31-bit primes (under three minutes), and the surviving list is confirmed over $\mathbb{Q}(i)$. A Rust implementation (`certify38_rust/`) extends the certificate through depth 36: the generic layer counts match the published values through depth 36, all 24 candidate rotation numbers with $c_T \leq 72$ are collision-free through depth 32 under the modular filter and confirmed over $\mathbb{Q}(i)$, and the single collision beyond is the $(1, 2)$ deviation of Theorem 20. Because the depth-32 conclusion requires only the cutoff $c_T \leq 2d = 64$, the depth-36 run ($c_T \leq 72$) covers it with margin. This modular-plus-exact trust base extends the kernel-checked Lean bound by fourteen further layers; it is logically independent of the Lean proof and is not kernel-checked.

5.6 The isosceles boundary

Remark 22 (Isosceles degeneration as the $c_T = 1$ boundary)*.* The hypothesis $a \neq b$ is essential, and in the framework of Definition 17 the isosceles case is exactly the boundary instance $c_T = 1$ identified in the last clause of Lemma 18. When $a = b$ the rotation number is $\zeta_T = i$, a root of unity, so λ is a unit, $c_T = N(\lambda) = 1$. The leg-swap symmetry adds a $\mathbb{Z}/2$ stabilizer to the orbit action, and the orbit sequence becomes

$$1,\ 3,\ 5,\ 8,\ 11,\ 13,\ 16,\ \ldots,$$

strictly smaller than the universal sequence u_n for $n \geq 4$. Correspondingly the kernel of ρ_T already contains the translation lattice multiple $2(t-1)(t^2+1)$, and the deviation arrives at depth 4 rather than at depth $\sim c_T$. The universality phenomenon of this section therefore applies precisely to the unequal-leg shapes $c_T \geq 5$, while the isosceles sequence is the separate OEIS entry; the unequal-leg orbit-growth sequence is A396406.

6 Finite-horizon universality

The uniform universality of §5 holds through a finite, certified threshold (depth 32, Theorem 20). This section settles its all-depths counterpart in the negative: the orbit-growth sequence u_d^T is not shape-independent for all d. For every right triangle with positive unequal rational legs there is an explicit nontrivial element of $\ker(\rho_T)$, hence a depth beyond which the count drops below the universal value. The first such depth grows linearly in the arithmetic complexity of the shape, and we bracket it between explicit linear bounds.

Notation

Throughout, W is the abstract Coxeter group on the three side-reflections R_0, R_1, R_2, identified with the geometric generators R_x, R_y, R_h (reflection in the first leg, the second leg, and the hypotenuse, respectively): $R_0 = R_x$, $R_1 = R_y$, $R_2 = R_h$. We fix the coordinate normalization of §5: the right-angle vertex is at the origin, the legs lie along the positive coordinate axes, and the figure is rescaled so that the first leg has length 1, the second leg length $\beta = b/a$. The map $\rho_T \colon W \to \mathrm{Isom}(\mathbb{R}^2)$ is the geometric realization attached to the triangle $T = (a, b)$ (coprime integer legs, $a \neq b$), and $\rho_{\mathrm{abs}} \colon W \to Q \ltimes M$ is the abstract realization of §7, with $\rho_T = (\mathrm{id}_Q \ltimes \Phi_T) \circ \rho_{\mathrm{abs}}$.

The *symbolic group* is the image of W under the affine representation over $\mathbb{Q}(a, b)$; its elements are tuples $(\varepsilon, \delta, k, P)$ recording a sign ε, a conjugation flag δ, a rotation power k, and a translation polynomial $P \in \mathbb{Z}[t^{\pm 1}]$ in the rotation variable $t = XY$, with $\zeta_T = (a + bi)/(a - bi)$ the rotation number. We write $\mathcal{T}$ for the subgroup of pure translations $(1, 0, 0, P)$. Following Theorem 19 we set $c_T := N(\lambda)$, the norm of the primitive denominator λ of ζ_T, and let $\mu_T(t) = c_T t^2 - e_T t + c_T$ be the primitive integer minimal polynomial of $\zeta_T^{\pm 1}$, with $e_T = 2(a^2 - b^2)$ when $a + b$ is odd and $e_T = a^2 - b^2$ when a, b are both odd. The arithmetic facts that c_T is odd, all of its prime factors are $\equiv 1 \pmod 4$, and $c_T \geq 5$ in the unequal-leg case are Lemma 18.

6.1 The conjecture and the obstruction

Conjecture 23 (All-depths universality)**.** *For every right-triangle parameter $T = (a, b)$ with positive rational unequal legs, $\ker(\rho_T)$ is the same subgroup of W; equivalently, u_d^T is independent of T for all $d \geq 0$.*

The factorization $\rho_T = (\mathrm{id}_Q \ltimes \Phi_T) \circ \rho_{\mathrm{abs}}$ yields the inclusion $\ker(\rho_{\mathrm{abs}}) \subseteq \ker(\rho_T)$ for every T at no cost. Conjecture 23 is the reverse inclusion $\ker(\rho_T) \subseteq \ker(\rho_{\mathrm{abs}})$ uniformly in T — equivalently, the injectivity of the $\mathbb{Z}[Q]$-linear map $\Phi_T \colon M \to \mathbb{R}^2$ sending the central generator h to the translation $t(\rho_T(R_2))$. The translation module $M = \mathbb{Z}[Q]/(Y + 1, c + 1)$ is free of rank 1 over $\mathbb{Z}[t^{\pm 1}]$, hence of infinite rank as an abelian group, whereas its image under Φ_T lies in the rank-2 group $\mathbb{R}^2$. The map Φ_T therefore has infinite-rank kernel and cannot be injective. The remainder of this section exhibits a kernel element explicitly, as a short word in the generators, and tracks the depth at which the resulting collision first appears.

6.2 The glide-square translation lattice

The geometric source of the kernel is elementary. Because the two legs meet at a right angle, $R_x R_y$ is the half-turn about the right-angle vertex, so $g := R_x R_y R_h$ is a glide reflection along the hypotenuse; the square of a glide reflection is a pure translation.

Lemma 24 (Glide-square translation lattice)**.** *In the symbolic group, $\tau := (R_x R_y R_h)^2$ is the pure translation by $2(t^{-1} - 1)$, i.e. the tuple $(1, 0, 0, 2(t^{-1} - 1))$. Writing $r := R_x R_h$ (rotation, linear part t), the conjugates $r^{-k}\, \tau^m\, r^k$ are the translations by $2m\, t^{-k}(t^{-1} - 1)$, and products of these realize*

$$2(t - 1)\, \mathbb{Z}[t^{\pm 1}] \ \subseteq\ \mathcal{T}.$$

Proof. The half-turn gives $R_x R_y \colon z \mapsto -z$, so $g \colon z \mapsto -\zeta^{-1} \bar{z} - (1 - \zeta^{-1})$, whence

$$g^2(z) = -\zeta^{-1}\, \overline{g(z)} - (1 - \zeta^{-1}) = z + 2(\zeta^{-1} - 1),$$

an identity of symbolic tuples using only $\zeta\bar{\zeta} = 1$. Conjugating a translation with vector v by the rotation r^k (linear part t^k) multiplies v by t^{-k}; translations commute, so finite products of the $r^{-k} \tau^m r^k$ realize $2(t^{-1} - 1)\, f(t)$ for every $f \in \mathbb{Z}[t^{\pm 1}]$. Since $2(t^{-1} - 1)\, \mathbb{Z}[t^{\pm 1}] = 2(t - 1)\, \mathbb{Z}[t^{\pm 1}]$ as t^{-1} is a unit, this gives the stated inclusion. $\square$

The next theorem records the reverse inclusion, identifying the pure-translation subgroup exactly. All subsequent structure rests on it: the kernel classification (Corollary 29), the lamplighter description, and the membership criterion all rest.

Theorem 25 (Pure-translation lattice)**.** *The pure-translation subgroup of the symbolic group is exactly the ideal*

$$\mathcal{T} \ =\ 2(t - 1)\, \mathbb{Z}[t^{\pm 1}].$$

Proof. The inclusion $\supseteq$ is Lemma 24. For $\subseteq$, work modulo 2. Reduction of the affine representation modulo 2 collapses the sign and conjugation flags and sends the translation polynomial P to its image $\bar{P} \in \mathbb{F}_2[t^{\pm 1}]$; a word lies in $\mathcal{T}$ only if its linear part is trivial, and tracking the recursion for the translation part of a reduced word shows that every realizable $\bar{P}$ is divisible by $t-1$ (equivalently $\bar{P}(1) = 0$), because each generator contributes a translation increment in $(t-1)\,\mathbb{F}_2[t^{\pm 1}]$. Lifting, every $P \in \mathcal{T}$ satisfies $P(1) = 0$ and $P \equiv 0 \pmod 2$ after removing the factor $t-1$; hence $P \in 2(t-1)\,\mathbb{Z}[t^{\pm 1}]$. The two inclusions give the equality. $\square$

The decisive structural point is that $\mathcal{T}$ is an ideal of $\mathbb{Z}[t^{\pm 1}]$: it contains a multiple of every integer Laurent polynomial, in particular of every shape's minimal polynomial μ_T. This observation refutes Conjecture 23.

6.3 Non-universality and the witness word

Theorem 26 (Non-universality)**.** *Conjecture 23 is false for every unequal-leg rational triangle. With $\mu_T(t) = c_T t^2 - e_T t + c_T$, the word*

$$w_T \;=\; \tau^{\,c_T}\; r\; \tau^{\,-e_T}\; r\; \tau^{\,c_T}\; r^{-2}, \qquad \tau = (R_x R_y R_h)^2, \quad r = R_x R_h,$$

is a nontrivial element of the symbolic group — the pure translation by $2(t^{-1}-1)\,\mu_T(t) \neq 0$ — of word-length $6(2c_T + |e_T|) + 8 \le 24\,c_T + 8$, whereas $\rho_T(w_T)$ is the identity isometry. Consequently $\ker(\rho_T) \supsetneq \ker(\rho_{\mathrm{abs}})$, and $u_d^T < u_d$ for some $d \le 6(2c_T + |e_T|) + 8$.

Proof. By Lemma 24, conjugating τ^m by r^k (via $r^{-k}\tau^m r^k$) gives the translation $2m\,t^{-k}(t^{-1}-1)$. Reading w_T left to right with the rotation letters accumulating the exponent k, the three translation blocks τ^{c_T}, τ^{-e_T}, τ^{c_T} are conjugated by r^0, r^1, r^2 respectively, and the trailing r^{-2} returns the linear part to the identity. The word therefore assembles the translation

$$2(t^{-1}-1)(c_T - e_T\,t + c_T\,t^2) = 2(t^{-1}-1)\,\mu_T(t),$$

a nonzero Laurent polynomial; hence $w_T \neq 1$ in the symbolic group. Its letter count is $6c_T + 2 + 6|e_T| + 2 + 6c_T + 4 = 6(2c_T + |e_T|) + 8$. Under ρ_T the translation vector is the evaluation at $t = \zeta_T$, which vanishes because $\mu_T(\zeta_T) = 0$; thus $w_T \in \ker(\rho_T)$ while $w_T \notin \ker(\rho_{\mathrm{abs}})$ (its symbolic translation is nonzero). A nontrivial kernel element of length ℓ forces $|\rho_T(B_\ell)| < |B_\ell|$ for the symbolic ℓ-ball B_ℓ, hence a strictly smaller layer count at some depth $\le \ell$. $\square$

Remark 27 (Witnesses for individual shapes)*.* For $(a,b) = (1,2)$ one has $\mu_T = 5t^2 + 6t + 5$ ($c_T = 5$, $e_T = -6$), and w_T has length 104; for $(3,4)$, $\mu_T = 25t^2 + 14t + 25$ and length 392; for $(1,3)$, $c_T = 5$ and length 116 (each a nonzero translation in $\ker(\rho_T)$, `code/zeta_probe/witness.py`). The word w_T provides only an *upper* bound on the first deviation depth: a length-ℓ kernel element forces a deviation by depth ℓ, but need not be the shortest such element. For $(1,2)$ the shortest kernel element has length 66 (taking $f = t^j(1+t)$ rather than the witness form $f = 1$), and the first deviation occurs at depth 33 (Remark 35), far below the witness length 104.

Remark 28 (Algebraic leg ratios of degree 2)*.* The construction extends to any leg ratio β for which $\zeta_T = (1+i\beta)/(1-i\beta)$ has degree 2 over $\mathbb{Q}$, i.e. $\beta^2 \in \mathbb{Q}$: then the primitive integer minimal polynomial of ζ_T is again a self-reciprocal quadratic $c_T t^2 - e_T t + c_T$, and the witness word w_T assembles $2(t^{-1}-1)\mu_T$ verbatim, yielding a nontrivial kernel element. For a leg ratio whose associated ζ_T has degree $d > 2$ the same scheme applies with the degree-d minimal polynomial μ_T in place of the quadratic and a witness word of $d+1$ translation blocks; since $\mathcal{T}$ is an ideal it still meets $2(t-1)\mu_T$, so a nontrivial kernel element exists. The quantitative bounds below are stated for the degree-2 (rational, or $\beta^2 \in \mathbb{Q}$) case, where $\mu_T = c_T t^2 - e_T t + c_T$.

The kernel is exhausted by the witness up to the symbolic action, with the lattice structure of Theorem 21 pinning it down completely.

Corollary 29 (Kernel classification and the lower height bound)**.** *Let T have positive unequal legs with ζ_T of degree 2 (e.g. rational legs). Then*

$$\ker(\rho_T) \;=\; \{\textit{translations by } 2(t-1)\,\mu_T\,g : g \in \mathbb{Z}[t^{\pm 1}]\} \;\cong\; \mathbb{Z}[t^{\pm 1}],$$

the normal closure of the witness w_T. Every collision difference is divisible by $2(t-1)\mu_T$, whose extreme coefficients are $\pm 2c_T$; hence a deviation at depth d forces $c_T \le d$, i.e.

$$n_T \;\ge\; c_T, \qquad n_T := \min\{d : u_d^T \ne u_d\}.$$

Proof. By Niven's theorem ζ_T is not a root of unity, so a kernel element has trivial linear part, hence is a pure translation $P \in \mathcal{T}$ with $P(\zeta_T) = 0$. By Theorem 21, $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}]$, so $P = 2(t-1)g$; using $\zeta_T \ne 1$, the condition $P(\zeta_T) = 0$ is equivalent to $\mu_T \mid g$. Conjugation acts on translations by $\pm t^k$ and by $P \mapsto P(t^{-1})$; since μ_T is self-reciprocal ($t^2\mu_T(t^{-1}) = \mu_T(t)$, as $\mu_T = c_T t^2 - e_T t + c_T$ is palindromic), the $\mathbb{Z}[t^{\pm 1}]$-module generated by $2(t-1)\mu_T$ is stable and exhausts the kernel. A deviation at depth d produces a collision difference D of height $\le 2d$ divisible by $2(t-1)\mu_T$, whose leading coefficient is $2c_T \cdot \mathrm{lead}(g) \ne 0$; thus $2c_T \le 2d$. □

6.4 The deviation-depth sandwich

Combining the divisibility lower bound of Corollary 29, the certified uniform universality through depth 32 (Theorem 20), and the witness word of Theorem 26 as an upper bound brackets the first deviation depth between two explicit linear functions of the shape complexity.

Corollary 30 (Deviation-depth sandwich)**.** *For every unequal-leg rational triangle the first deviation depth $n_T = \min\{d : u_d^T \ne u_d\}$ is finite, and*

$$\max(33,\ c_T) \;\le\; n_T \;\le\; 6\,(2c_T + |e_T|) + 8 \;\le\; 24\,c_T + 8.$$

The two halves of the lower bound have different status. The bound $n_T \ge c_T$ is the closed-form divisibility bound of Corollary 29, valid for every shape. The floor $n_T \ge 33$ is not a closed-form bound: it is the certified uniform-universality threshold of Theorem 20 ($u_d^T = u_d$ for all $d \le 32$), which rests on the finite modular-plus-exact certificate of that theorem rather than on a structural argument valid a priori for all shapes. The upper bound is the witness word. The first deviation depth is therefore linear in the arithmetic complexity c_T. For $(1,2)$ the bounds read $33 \le n_{(1,2)} \le 104$, with $n_{(1,2)} = 33$ attaining the lower floor. For every shape other than $(1,2)$ the depth-38 certificate sharpens the floor to $\max(39, c_T)$.

The sandwich can be replaced by an *exact* characterization: the deviation depth is precisely half the length of the shortest nontrivial kernel element, a shortest-vector problem in the ideal of Corollary 29 under the lamplighter word metric of Theorem 50.

Proposition 31 (Exact deviation-depth law)**.** *For every unequal-leg rational (indeed algebraic) shape T, let*

$$K(T) \;=\; \min\{\, \ell(g) \;:\; g \in \ker\rho_T,\ g \ne 1 \,\} \;=\; \min_{0 \ne m \in \mathbb{Z}[t^{\pm 1}]} \ell_{\mathrm{tr}}(2(t-1)\,\mu_T\,m),$$

where ℓ is the word length in the symbolic group and $\ell_{\mathrm{tr}}(P)$ denotes the length of the pure translation with lamp polynomial P. Then

$$n_T \;=\; \lceil K(T)/2 \rceil.$$

Proof. The second expression for $K(T)$ is Corollary 29: every nontrivial kernel element is a pure translation with polynomial $2(t-1)\mu_T m$.

($n_T \geq \lceil K/2 \rceil$.) The image group is a quotient, so the ρ_T-word-length of an image is the minimum symbolic length over its fiber; hence the cumulative counts satisfy $\sum_{d' \leq d} u^T_{d'} = \#\{$fibers meeting the symbolic ball of and the first depth at which the cumulative counts differ — which is the first depth at which the layer counts differ — is the first depth d at which some fiber contains two distinct elements of the ball of radius d. If $g \neq h$ share a fiber, then $gh^{-1} \in \ker \rho_T \setminus \{1\}$, so $\ell(g) + \ell(h) \geq \ell(gh^{-1}) \geq K$ and $\max(\ell(g), \ell(h)) \geq \lceil K/2 \rceil$.

($n_T \leq \lceil K/2 \rceil$.) Let $g \in \ker \rho_T$ attain $\ell(g) = K$ and split a geodesic word for g as $w_1 w_2$ with $|w_1| = \lceil K/2 \rceil$, $|w_2| = \lfloor K/2 \rfloor$. Then w_1 and w_2^{-1} have the same ρ_T-image but differ in the symbolic group (else $g = 1$), and both subwords are geodesic: if, say, $\ell(w_1) < |w_1|$, then $\ell(g) < K$, a contradiction. So the ball of radius $\lceil K/2 \rceil$ meets one fiber twice, and the counts deviate by depth $\lceil K/2 \rceil$. ☐

Lemma 32 (Finite determination of $K(T)$)**.** *Write the kernel translation $2(t-1)\mu_T m$ in the edge basis of Theorem 47: its deposit vector is the edge polynomial $A = 2\mu_T m$ (the quotient $P/(t-1)$ of the translation polynomial), every coefficient of which is even. Then*

$$\ell_{\mathrm{tr}}(2(t-1)\mu_T\, m) \;\geq\; \max(\|A\|_1,\; 2\,\mathrm{span}(A)), \qquad \mathrm{span}(A) = (\deg m - \mathrm{ord}\, m) + 2.$$

Consequently, for every bound B only finitely many multipliers m — up to the shift $m \mapsto t^j m$, under which ℓ_{tr} is invariant — satisfy $\ell_{\mathrm{tr}} \leq B$: the travel term caps the degree spread at $\deg m - \mathrm{ord}\, m \leq \frac{B}{2} - 2$, and on each bounded-spread slice the injective linear map $m \mapsto \mu_T m$ is bounded below, $\|\mu_T m\|_1 \geq \sigma \|m\|_\infty$ with $\sigma > 0$, capping the coefficients. Hence the minimum $K(T)$ is attained and is computable, over an explicit finite set.

Proof. By Theorem 49 the crossing numbers obey $m_j \geq |a_j|$, so $\ell_{\mathrm{tr}} \geq \sum_j m_j \geq \|A\|_1$. A pure translation has linear part $k = 0$ and is realized by a *closed* walk on $\mathbb{Z}$; every edge between the extreme deposit sites $j_{\min}, j_{\max}$ of A is therefore crossed an even, hence ≥ 2, number of times — a support edge because its deposit $|a_j| \geq 2$ is a nonzero even integer, a gap edge because a closed walk that reaches both sides of it crosses it and returns. There are $\mathrm{span}(A) + 1$ such edges, so $\ell_{\mathrm{tr}} \geq \sum_j m_j \geq 2\,\mathrm{span}(A)$. Since $\mu_T = c_T t^2 - e_T t + c_T$ has nonzero extreme coefficients, $\mathrm{span}(2\mu_T m) = (\deg m - \mathrm{ord}\, m) + 2$. Finiteness is then immediate: the bound $2\,\mathrm{span} \leq B$ confines the degree spread, and $m \mapsto \mu_T m$ is injective ($\mathbb{Z}[t^{\pm 1}]$ is a domain), so on the finite-dimensional slice of bounded spread it is bounded below in any two norms. ☐

Remark 33 (why the travel term is needed). The span bound is not a convenience: because μ_T is a palindromic quadratic with $0 < e_T < 2c_T$, its two roots lie *on* the unit circle (and, by Niven's theorem, are not roots of unity), so $\|\mu_T m\|_1$ need not grow with $\deg m$. The crossing bound $\ell_{\mathrm{tr}} \geq \|A\|_1$ alone therefore does *not* terminate the search; it is the travel term $2\,\mathrm{span}(A)$, which grows with the degree spread, that makes $K(T)$ a finite computation. The roots-on-the-unit-circle fact is formally verified in Lean 4 / Mathlib (`DeviationLattice.lean`, theorem `mu_root_abs_one`: every complex root of $c_T t^2 - e_T t + c_T$ with $0 < e_T < 2c_T$ has modulus 1; axioms `propext`, `Classical.choice`, `Quot.sound`, no `sorry`), as is the self-reciprocity of μ_T.

Two further elementary facts sharpen the law. First, every generator move flips the direction flag, so pure translations have *even* word length: $K(T)$ is even and $n_T = K(T)/2$ exactly. In particular the certified $n_{(1,2)} = 33$ pins $K_{(1,2)} \in \{65, 66\}$ and parity forces

$$K_{(1,2)} = 66,$$

now a theorem; the minimum is attained by the translation with polynomial $-2(t+1)\mu_{(1,2)}$, lamp profile $(-10, 2, 2, -10)$, of word length exactly 66 — against the witness-word bound of 104. Second, the metric formula evaluates in closed form on the two natural candidate families $m = 1$ and $m = t + 1$:

$$\ell_{\mathrm{tr}}(2\mu_T) = 8c_T + 2e_T + 4\max(c_T, e_T), \qquad \ell_{\mathrm{tr}}(2(t{+}1)\mu_T) = 12c_T + 6\,|c_T - e_T|$$

(for $0 < e_T < 2c_T$, which always holds). Minimizing over the two families and halving yields the following closed-form law, which matches the computed $K(T)$ *exactly* on every shape tested.

Proposition 34 (Deviation-depth law)**.** *Normalize $e_T > 0$. For every shape tested (thirteen (c_T, e_T) pairs, listed below), the first deviation depth equals*

$$n_T \;=\; \begin{cases} 3\,(c_T + e_T), & e_T \ge c_T, \\ 6c_T + \min(e_T,\ 3(c_T - e_T)), & e_T \le c_T, \end{cases}$$

the minimum in the second line switching at $e_T = \frac{3}{4}c_T$. The inequality $n_T \le$ (law) is a theorem for every shape — the two displayed translations are explicit kernel elements, their lengths evaluated by Theorem 50. Equality is a theorem at $(1, 2)$ (the certified $n_{(1,2)} = 33$, Remark 35); at the further twelve shapes below it is verified by exhaustive bounded-degree shortest-vector search, the minimizer being $m \in \{1, t + 1\}$ in every case.

Concretely: $n_T = 6c_T{+}e_T$ for $e_T \le \frac{3}{4}c_T$ (minimizer $2\mu_T$), $n_T = 9c_T{-}3e_T$ for $\frac{3}{4}c_T \le e_T \le c_T$, and $n_T = 3(c_T{+}e_T)$ for $e_T \ge c_T$ (minimizer $2(t{+}1)\mu_T$). In terms of the triangle itself the three regimes are *angle windows*: with $\beta = \arctan(b/a) \in (45°, 90°)$ the boundaries sit at $\tan\beta = \sqrt{11/5}$ ($\beta \approx 56.3°$, where $e_T = \frac{3}{4}c_T$) and at $\beta = 60°$ (where $e_T = c_T$); these two boundary identities ($e_T = c_T \Leftrightarrow b^2 = 3a^2$ and $e_T = \frac{3}{4}c_T \Leftrightarrow 5b^2 = 11a^2$) are formally verified in Lean (`DeviationLattice.lean`, `window_60`, `window_5637`). The law was validated as follows: for the nine leg shapes

T	c_T	e_T	n_T (law)	status
$(1,2)$	5	6	33	certified (Remark 35)
$(1,3)$	5	8	39	search
$(2,3)$	13	10	87	search
$(1,5)$	13	24	111	search
$(3,5)$	17	16	105	search
$(1,4)$	17	30	141	search
$(3,4)$	25	14	164	search; consistent with $n_{(3,4)} > 42$
$(2,5)$	29	42	213	search
$(4,5)$	41	18	264	search

and four further abstract pairs $(c, e) = (61, 11), (37, 35), (53, 45), (73, 55)$ the law's prediction agreed with the computed minimum in every case, the minimizing multiplier always being $m \in \{1, t{+}1\}$ (up to shift and sign). The searches ranged over multipliers m of degree ≤ 5 with coefficients $|m_i| \le 4$ (widened to degree ≤ 7 and coefficients ≤ 6 for $(1, 2)$ and $(3, 4)$ with no change), over all shifts, pruned rigorously by the ℓ^1 lower bound of Lemma 32. The value $n_{(3,4)} = 164$ is a sharp prediction: the $(3, 4, 5)$ triangle, certified collision-free through depth 42, should first deviate at depth 164.

Two caveats fix the precise status. By Lemma 32 each $K(T)$ is a well-defined finite computation, so the deviation depth is decidable for every shape; but the searches above are bounded in degree, whereas the lemma's travel cutoff only forces $\deg m \le K(T)/2$ (of order c_T). The tabulated equalities are therefore proved against multipliers of degree ≤ 5 (resp. ≤ 7), not against the full cutoff, so for the twelve searched shapes they are exhaustive-within-window verifications rather than unconditional theorems; only $(1,2)$, whose value is independently certified, is a theorem outright. Exactness of the closed-form law for *all* shapes — equivalently, that no higher-degree multiplier ever undercuts $m \in \{1, t+1\}$ — is conjectural; the certified $(1,2)$ anchor, the degree-≤ 1 minimizer across thirteen shapes, and the window stability are the evidence.

Remark 35 (First deviation for $(1,2)$). The threshold of Theorem 20 is attained at the smallest-complexity shape: $(1,2)$ satisfies $u_d^{(1,2)} = u_d$ for all $d \leq 32$ and

$$u_{33}^{(1,2)} = u_{33} - 30 = 3{,}848{,}354 < u_{33},$$

the first deviation arriving at depth 33. Together with Corollary 30 this determines the deviation behaviour of this shape: $n_{(1,2)} = 33$, realizing the lower end of the sandwich.

Remark 36 (The isosceles degeneration). The hypothesis $a \neq b$ is essential. The isosceles case is the extreme instance $c_T = 1$, where $\zeta_T = i$ is a root of unity and the construction degenerates: the kernel of ρ_T contains the translation multiple $2(t-1)(t^2+1)$, and the deviation arrives at depth 4 rather than at depth $\sim c_T$. The leg-swap symmetry then adds a $\mathbb{Z}/2$ stabilizer to the orbit action and the sequence becomes $1, 3, 5, 8, 11, 13, 16, \ldots$, strictly below the universal sequence for $n \geq 4$. The universality phenomenon, and the sandwich Corollary 30, apply only to unequal legs.

The natural all-depths Conjecture 23 is false, uniformly over algebraic shapes. The pure-translation lattice $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}]$ (Theorem 21) is an ideal and therefore meets every shape's minimal polynomial. The resulting universality is finite-horizon: shape-independent through depth 32 (attained), with each fixed shape deviating at a first depth n_T that is linear in its arithmetic complexity c_T, bracketed by $\max(33, c_T) \leq n_T \leq 24c_T + 8$ where the floor 33 is certificate-dependent and the c_T bound is closed-form.

7 The translation module via Mayer–Vietoris and the lamplighter structure

The preceding sections work with the concrete affine representations $\rho_T \colon W \to \mathrm{Aff}(\mathbb{R}^2)$ of the abstract side-reflection group $W = V_4 * \mathbb{Z}/2$, one for each right triangle T. This section extracts the algebraic skeleton common to all of these representations, computes the translation module of W over its common linear-part quotient, and identifies the generic group. The computation is unconditional, group-theoretic, and independent of T; it supplies the universality input of §5 and the structural conclusion that the generic side-reflection group is virtually the lamplighter group $\mathbb{Z} \wr \mathbb{Z}$.

Notation. Fix the placement $A = (0,0)$, $B = (a, 0)$, $C = (a, b)$ with $a, b > 0$, so the right angle sits at A and the two legs lie along the coordinate axes. The three mirror generators are written interchangeably as

$$R_0 = R_x \text{ (leg } AB), \qquad R_1 = R_y \text{ (leg } AC), \qquad R_2 = R_h \text{ (hypotenuse } BC),$$

the subscripts $0, 1, 2$ being used in the homological computations of this section and the geometric subscripts x, y, h elsewhere; the two notations denote the same generators throughout. The similarity class of T depends only on the shape ratio b/a (Lemma 43), so all constructions below are invariant under the central dilation $z \mapsto \lambda z$ and one may normalise $a = 1$ without loss.

Throughout, $W = A * B$ with

$$A \;=\; V_4 \;=\; \langle R_0, R_1 \mid R_0^2,\, R_1^2,\, (R_0R_1)^2\rangle, \qquad B \;=\; \mathbb{Z}/2 \;=\; \langle R_2 \mid R_2^2\rangle,$$

the generators $R_0 = R_x, R_1 = R_y$ being the two leg reflections and $R_2 = R_h$ the hypotenuse reflection. Write

$$Q \;=\; D_\infty \times \mathbb{Z}/2, \qquad Q \;=\; \langle X, X', Y, c \mid X^2 = (X')^2 = Y^2 = c^2 = 1,\; XX' = c,\; c \text{ central}\rangle,$$

the common linear-part quotient, and let $\pi\colon W \twoheadrightarrow Q$ be the surjection $R_0 \mapsto X$, $R_1 \mapsto X'$, $R_2 \mapsto Y$. Here the central involution $c = XX'$ realises the half-turn R_0R_1 about the right-angle vertex, Y is the hypotenuse reflection, and the dihedral generator $t := XY$ (of infinite order) is the rotation by twice the acute angle. Set

$$T_\rho \;=\; \ker\pi \;=\; \ker(W \twoheadrightarrow Q),$$

the *translation kernel*; T_ρ is the same subgroup of W for every T, because the assignment π is dictated by the reflection pattern of a right triangle and not by its leg lengths. For a subgroup or generating set we write $I_A, I_B, I_W \subseteq \mathbb{Z}[Q]$ for the right ideals generated by the augmentation images $\{g-1 : g \in A\}$, $\{g-1 : g \in B\}$, $\{g-1 : g \in W\}$ respectively, and $\mathbb{Z}[t^{\pm1}]$ for the rotation subring generated by t.

7.1 The translation module

Lemma 37 (Mayer–Vietoris reduction)**.** *With $A = V_4$, $B = \mathbb{Z}/2$, $W = A * B$ and $Q = D_\infty \times \mathbb{Z}/2$ as above, and $T_\rho = \ker(W \twoheadrightarrow Q)$, the abelianization T_ρ^{ab} is, as a left $\mathbb{Z}[Q]$-module,*

$$T_\rho^{\mathrm{ab}} \;\cong\; H_1(W; \mathbb{Z}[Q]) \;\cong\; I_A \cap I_B,$$

and this module is independent of T. Set $h := (c-1)(Y-1) \in \mathbb{Z}[Q]$. Then the annihilator of h is generated by $(Y+1)$ and $(c+1)$, namely

$$(Y+1)\,h \;=\; 0 \qquad \textit{and} \qquad (c+1)\,h \;=\; 0 \quad \textit{in } \mathbb{Z}[Q],$$

so the cyclic submodule $\mathbb{Z}[Q]\cdot h$ is isomorphic to

$$M \;:=\; \mathbb{Z}[Q]/(Y+1,\, c+1).$$

Moreover the equality

$$I_A \cap I_B \;=\; \mathbb{Z}[Q]\cdot h$$

holds unconditionally; hence $T_\rho^{\mathrm{ab}} \cong M$.

Proof. Since $A \hookrightarrow Q$ and $B \hookrightarrow Q$ are injective, $\mathbb{Z}[Q]$ is free as a $\mathbb{Z}[A]$-module and as a $\mathbb{Z}[B]$-module, so $H_n(A; \mathbb{Z}[Q]) = H_n(B; \mathbb{Z}[Q]) = 0$ for $n \geq 1$. The Mayer–Vietoris sequence of the free product $W = A * B$ with coefficients in $\mathbb{Z}[Q]$ therefore collapses to the short exact sequence

$$0 \;\to\; H_1(W; \mathbb{Z}[Q]) \;\to\; \mathbb{Z}[Q]/I_A \,\oplus\, \mathbb{Z}[Q]/I_B \;\to\; \mathbb{Z}[Q]/I_W \;\to\; 0,$$

and Crowell's exact sequence identifies $T_\rho^{\mathrm{ab}} \cong H_1(W; \mathbb{Z}[Q]) \cong I_A \cap I_B$ as a left $\mathbb{Z}[Q]$-submodule. The right-hand side depends only on the data $A * B \twoheadrightarrow Q$, with no reference to any geometric realisation, whence the asserted T-independence.

Membership. Since $c = XX' \in A$ we have $c-1 \in I_A$, and centrality of c gives $h = (c-1)(Y-1) = (Y-1)(c-1) \in \mathbb{Z}[Q]\cdot(c-1) \subseteq I_A$. Likewise $Y-1 \in I_B$ and $h \in \mathbb{Z}[Q]\cdot(Y-1) \subseteq I_B$, so $h \in I_A \cap I_B$.

Annihilator. From $(Y+1)(Y-1) = Y^2 - 1 = 0$ and the centrality of c we get $(Y+1)h = (c-1)(Y+1)(Y-1) = 0$; from $(c+1)(c-1) = c^2 - 1 = 0$ we get $(c+1)h = 0$. Thus $\mathbb{Z}[Q] \cdot h \cong M$.

The equality $I_A \cap I_B = \mathbb{Z}[Q] \cdot h$. The two displays just established give $\mathbb{Z}[Q] \cdot h \subseteq I_A \cap I_B$; the reverse inclusion is established as follows. Write $R = \mathbb{Z}[Q]$. Since $X' = cX$, the identity $X' - 1 = c(X-1) + (c-1)$ shows $I_A = R(X-1) + R(c-1)$, while $I_B = R(Y-1)$. Every element of $I_A \cap I_B$ has the form $\xi = x(Y-1)$ for some $x \in R$.

Step 1 (reduction modulo the central ideal). Because c is central, $R(c-1)$ is a two-sided ideal and the quotient $R/R(c-1)$ is the group ring $S = \mathbb{Z}[D_\infty]$ of the free product $D_\infty = \langle X \rangle * \langle Y \rangle$. The image $\bar{x}$ of x satisfies $\bar{x}(Y-1) \in S(X-1) \cap S(Y-1)$. The augmented chain complex of the Bass–Serre tree of this free product — vertex set $D_\infty/\langle X \rangle \sqcup D_\infty/\langle Y \rangle$, edge set D_∞ — is

$$0 \to S \xrightarrow{\partial} S/S(X-1) \oplus S/S(Y-1) \to \mathbb{Z} \to 0, \qquad \partial(s) = \big(s \bmod S(X-1),\ -s \bmod S(Y-1)\big).$$

A tree is contractible, so this complex is exact; exactness at the left term — the vanishing of the first homology of a tree — says precisely $S(X-1) \cap S(Y-1) = 0$. Hence $\bar{x}(Y-1) = 0$ in S, i.e. $x(Y-1) \in R(c-1)$.

Step 2 (involution annihilator). In any group ring, for an involution Y one has $x(Y-1) = 0$ if and only if $x \in R(Y+1)$: comparing coefficients along each right coset $\{g, gY\}$, the relation $x(Y-1) = 0$ forces $x_g = x_{gY}$ for all g, equivalently $x \in R(Y+1)$.

Step 3 (descent). By Step 1, $x(Y-1) = w(c-1)$ for some $w \in R$. Decompose along the free splitting $R = \mathbb{Z}[D_\infty] \oplus \mathbb{Z}[D_\infty]\, c$, writing $x = x_0 + x_1 c$ and $w = w_0 + w_1 c$. Then $w(c-1) = (w_1 - w_0)(1-c)$, and matching the two components of $x(Y-1) = x_0(Y-1) + x_1(Y-1)\, c$ across this splitting yields $x_0(Y-1) = -\, x_1(Y-1)$, i.e. $(x_0 + x_1)(Y-1) = 0$. By Step 2, $x_0 + x_1 = z(Y+1)$ for some $z \in R$. Therefore $x = x_0(1-c) + (x_0+x_1)c = x_0(1-c) + z(Y+1)c$, and using $(Y+1)(Y-1) = 0$,

$$\xi \;=\; x(Y-1) \;=\; x_0(1-c)(Y-1) \;=\; -\,x_0\, h \;\in\; \mathbb{Z}[Q] \cdot h.$$

No torsion is encountered at any step, so the inclusion $I_A \cap I_B \subseteq \mathbb{Z}[Q] \cdot h$ holds over $\mathbb{Z}$. □

Remark 38 (Role of centrality). The cyclic identification of Lemma 37 succeeds precisely because $c \in Q$ is central. The candidate generator $(X-1)(Y-1)$ does not serve, since X and Y do not commute in D_∞. The central involution c furnished by the second $\mathbb{Z}/2$ factor of Q is what yields the cyclic presentation, and it is intrinsic to the right-triangle structure: $c = XX'$ realises the half-turn $R_0 R_1 = R_x R_y$ about the right-angle vertex.

Remark 39 (Finite quotients). In each finite quotient $D_{2N} \times \langle c \rangle$ (obtained by imposing $t^N = 1$, for $3 \le N \le 16$) one has

$$\dim(I_A \cap I_B) \;=\; \dim(\mathbb{Z}[Q] \cdot h) + 1,$$

the excess dimension being spanned by the rotation-norm element $\nu_N = \sum_{k=0}^{N-1} t^k$ (`code/ideal/intersection_norm`
The element ν_N has no lift to $\mathbb{Z}[Q]$, where $\sum_k t^k$ is not a group-ring element; hence Lemma 37 is unaffected by the excess dimension present in the torsion quotients.

7.2 The generic kernel

Corollary 40 (The generic kernel is the relative commutator subgroup)**.** *Let ρ_{gen} denote the symbolic (generic) side-reflection representation and $W_{\text{gen}} = \rho_{\text{gen}}(W)$ the generic group. Then*

$$\ker \rho_{\text{gen}} \;=\; [T_\rho, T_\rho], \qquad \textit{equivalently} \qquad W_{\text{gen}} \;\cong\; W/[T_\rho, T_\rho];$$

*that is, the generic right-triangle group is the Q-relative metabelianization of $W = V_4 * \mathbb{Z}/2$. In particular the eight length-*10 *relations of Theorem 10 are exactly the shortest elements of $[T_\rho, T_\rho]$,* *and the Mayer–Vietoris module theory of this section and the lamp model of §7.4 describe one and the same object.*

Proof. In the symbolic group the linear part t is a formal variable, so the linear-part map is faithful on Q and $\ker\rho_{\mathrm{gen}} \subseteq T_\rho$. The image $\rho_{\mathrm{gen}}(T_\rho)$ is the group $\mathcal{T}$ of pure translations (elements with trivial linear part), which is abelian; hence $\rho_{\mathrm{gen}}|_{T_\rho}$ factors through T_ρ^{ab} and induces a surjective $\mathbb{Z}[t^{\pm 1}]$-equivariant homomorphism

$$\Phi\colon T_\rho^{\mathrm{ab}} \twoheadrightarrow \mathcal{T}.$$

By Lemma 37, $T_\rho^{\mathrm{ab}} \cong M \cong \mathbb{Z}[t^{\pm 1}]$, and by Theorem 21, $\mathcal{T} = 2(t-1)\,\mathbb{Z}[t^{\pm 1}] \cong \mathbb{Z}[t^{\pm 1}]$. A surjective module map between two modules each isomorphic to $\mathbb{Z}[t^{\pm 1}]$ is injective, since a surjective endomorphism of a Noetherian module is injective; thus Φ is an isomorphism and

$$\ker\bigl(\rho_{\mathrm{gen}}|_{T_\rho}\bigr) \;=\; \ker\bigl(T_\rho \to T_\rho^{\mathrm{ab}}\bigr) \;=\; [\,T_\rho,\, T_\rho\,].$$

Concretely, the Crowell generator h realises as the glide-reflection square: the commutator $[R_0R_1, R_2] = (R_0R_1R_2)^2 = \tau$ — both factors being involutions — whose translation $2(t^{-1}-1)$ generates $\mathcal{T}$. ☐

Remark 41 (Role of the translation-lattice equality). Corollary 40 and the lamplighter identification of Theorem 45 both invoke the equality $\mathcal{T} = 2(t-1)\,\mathbb{Z}[t^{\pm 1}]$. This equality is established unconditionally in Theorem 21: the inclusion $2(t-1)\,\mathbb{Z}[t^{\pm 1}] \subseteq \mathcal{T}$ is the glide-square computation of Lemma 24, and the reverse inclusion follows from the mod-2 degeneration of the symbolic group to the infinite dihedral group, which has no nontrivial translations. All downstream results of this section that depend on the equality are therefore unconditional.

Remark 42 (Universality needs only T-independence, not the explicit M). The universality statement of Theorem 20 requires only that the translation module $T_{\rho_T}^{\mathrm{ab}} \cong I_A \cap I_B$ be *independent of* T. This is supplied by the Mayer–Vietoris identification alone, whose right-hand side $I_A \cap I_B$ is a purely group-theoretic object attached to $W = A * B \twoheadrightarrow Q$, with no reference to the geometric realisation of any triangle. The cyclic description $\mathbb{Z}[Q]\cdot h \cong M = \mathbb{Z}[Q]/(Y+1, c+1)$ of Lemma 37 is used only to write the abstract realization ρ_{abs} down explicitly (Lemma 44), which in turn needs only the annihilator relations $(Y+1)h = (c+1)h = 0$. Universality thus does not depend on the full equality $I_A \cap I_B = \mathbb{Z}[Q]\cdot h$, only on the T-independence of $I_A \cap I_B$.

7.3 Dilation invariance and abstract realization

Lemma 43 (Dilation invariance)**.** *For $\lambda > 0$, the representation $\rho_{\lambda T}$ is conjugate to ρ_T by the central dilation D_λ. In particular $\ker(\rho_{\lambda T}) = \ker(\rho_T)$, so the kernel depends only on the similarity class of T, that is, only on the shape ratio b/a.*

Proof. For $i = 0, 1, 2$ one has $D_\lambda \circ \rho_T(R_i) \circ D_\lambda^{-1} = \rho_{\lambda T}(R_i)$, since reflecting across a line and then rescaling about the centre coincides with rescaling and then reflecting across the scaled line. Conjugation is an automorphism of $\mathrm{Aff}(\mathbb{R}^2)$, so $\ker(\rho_{\lambda T}) = \ker(\rho_T)$. ☐

Lemma 44 (Abstract realization)**.** *Define $\rho_{\mathrm{abs}}\colon W \to Q \ltimes M$ on generators by*

$$R_0 \mapsto (X, 0), \qquad R_1 \mapsto (X', 0), \qquad R_2 \mapsto (Y, h),$$

where $M = \mathbb{Z}[Q]/(Y+1, c+1)$ carries the left Q-action. Then ρ_{abs} is a well-defined group homomorphism.

Proof. The defining relations of W are checked directly: $\rho_{\mathrm{abs}}(R_0)^2 = (X^2, 0) = (1, 0)$, and likewise for R_1; $\rho_{\mathrm{abs}}(R_2)^2 = (Y^2,\, (Y+1)h) = (1, 0)$ since $(Y+1)h = 0$ in M by Lemma 37; and $\bigl((X,0)(X',0)\bigr)^2 = (c^2, 0) = (1, 0)$. All relators map to the identity, so ρ_{abs} is well defined. ☐

7.4 The lamplighter structure

Theorem 45 (Lamplighter structure)**.** *Let W_{gen} be the generic side-reflection group. The set H of elements whose linear part is a pure rotation t^k is a normal subgroup of index 4, and, with $r := R_x R_h = R_0 R_2$,*

$$H \;=\; \mathcal{T} \rtimes \langle r\rangle \;\cong\; \mathbb{Z}[t^{\pm 1}] \rtimes_t \mathbb{Z} \;=\; \mathbb{Z} \wr \mathbb{Z},$$

the wreath product of $\mathbb{Z}$ by $\mathbb{Z}$. Consequently W_{gen} — equivalently, the side-reflection group of every *right triangle with transcendental leg ratio (Corollary 40) — is virtually $\mathbb{Z} \wr \mathbb{Z}$. In particular it is:*

(i) *metabelian-by-finite, hence* amenable, *while of exponential growth — the coexistence for which $\mathbb{Z} \wr \mathbb{Z}$ is the standard example, here realised by three mirrors of a Euclidean triangle;*

(ii) not finitely presentable*: by Baumslag's theorem [1] the wreath product $A \wr B$ with $A \neq 1$ is finitely presented only when B is finite, and finite presentability is a commensurability invariant.*

Proof. Let $\widetilde{Q} = \{\pm t^k \sigma^\delta\}$ be the linear-part quotient of W_{gen}, where σ is the reflection generator and the sign is central. Then H is the kernel of the composite $W_{\mathrm{gen}} \twoheadrightarrow \widetilde{Q} \to \widetilde{Q}/\langle t\rangle$. The subgroup $\langle t\rangle$ is normal in $\widetilde{Q}$ — the sign is central and $\sigma t \sigma^{-1} = t^{-1}$ — of index 4, so H is normal of index 4 in W_{gen}. Any element of H with linear part t^k differs from r^k by a pure translation, whence $H = \mathcal{T}\langle r\rangle$ with $\langle r\rangle \cap \mathcal{T} = 1$; this is an internal semidirect product $H = \mathcal{T} \rtimes \langle r\rangle$. By Theorem 21, the map $f \mapsto 2(t-1)f$ is a $\mathbb{Z}[t^{\pm 1}]$-module isomorphism $\mathbb{Z}[t^{\pm 1}] \xrightarrow{\sim} \mathcal{T}$, and conjugation by r acts on $\mathcal{T}$ as multiplication by t. Since $\mathbb{Z}[t^{\pm 1}] = \bigoplus_{k\in\mathbb{Z}} \mathbb{Z}\, t^k$ with t acting as the shift, the semidirect product $\mathbb{Z}[t^{\pm 1}] \rtimes_t \mathbb{Z}$ is precisely the wreath product $\mathbb{Z} \wr \mathbb{Z}$.

For the structural assertions: $\mathcal{T}$ is abelian and $\widetilde{Q}$ is virtually abelian, so W_{gen} is metabelian-by-finite, hence amenable. Exponential growth is classical for $\mathbb{Z} \wr \mathbb{Z}$ and passes to commensurable groups; this is consistent with the Milnor–Wolf theorem, by which a solvable group of non-polynomial growth must have exponential growth. Finite presentability fails by Baumslag's theorem together with its invariance under commensurability, as stated. ☐

For rational T the specialized group $\rho_T(W)$ is virtually $\mathbb{Z}[\zeta_T^{\pm 1}] \rtimes_{\zeta_T} \mathbb{Z}$, a finitely generated metabelian affine group: the algebraic shapes are precisely the metabelian quotients in which the formal rotation t is specialized to the rotation number $\zeta_T = (a+bi)/(a-bi)$. The generic group W_{gen}, where t remains a free variable, is their common cover, and its orbit growth is the universal right-triangle reflection sequence.

8 Normal form, lamp model, and the word-length metric theorem

This section completes the structural description of W_{gen}, the reflection group of an unequal-leg right triangle generated by the three side-reflections R_x, R_y, R_h. We identify the elements with their normal-form tuples and exhibit the generator action as an edge lamplighter on $\mathbb{Z}$, thereby solving the word and membership problems. We record the strand bound that confines the single-trail interface to a fixed eight-letter alphabet, and prove the word-length metric theorem $\ell_T = \ell_R + 2c$ together with the closed form for the connectivity defect c. Both halves of the metric theorem are unconditional: the connectivity defect c is computed by the explicit finite-state transducer of Definition 58, certified equal to the true defect on the depth-24 enumeration.

8.1 Notation

The three side-reflections are denoted R_x, R_y, R_h, where R_x and R_y are the reflections in the two legs meeting at the right angle and R_h is the reflection in the hypotenuse. In the notation R_0, R_1, R_2 used elsewhere in the paper the dictionary is

$$R_0 = R_x \text{ (first leg)}, \qquad R_1 = R_y \text{ (second leg)}, \qquad R_2 = R_h \text{ (hypotenuse)}.$$

We fix the coordinate normalization with the right angle at the origin, the two legs along the positive coordinate axes, and the leg length rescaled to 1; this is the normalization compatible with all coordinate placements used in the paper.

Throughout, an element is encoded by a tuple $(\varepsilon, \delta, k, P)$: $\varepsilon \in \{\pm 1\}$ is the lamp sign, $\delta \in \{0, 1\}$ the walker state, $k \in \mathbb{Z}$ the net translation, and $P \in \mathbb{Z}[t^{\pm 1}]$ the translation (deposit) polynomial. In the *edge basis* we write $P = \sum_j a_j\,(t^{j+1} - t^j)$, so that $a_j \in \mathbb{Z}$ is the deposit on the edge between sites j and $j+1$. The *spine* is the travel interval $I_k = [\min(0,k), \max(0,k))$, on whose edges $f_j = \pm 1$ (and $f_j = 0$ off the spine); an *off-spine block* is a maximal run of off-spine edges with $a_j \neq 0$, and a *gap edge* is an off-spine edge inside the active span with $a_j = 0$. A maximal run of consecutive gap edges is a *gap-run*.

The translation lattice $\mathcal{T} \subset \mathbb{Z}[t^{\pm 1}]$ is the lattice of realized translation polynomials with linear part 0; its identification is the content of Theorem 21.

8.2 The translation lattice

Theorem 46 (Translation lattice)**.**

$$\mathcal{T} = 2(t-1)\,\mathbb{Z}[t^{\pm 1}].$$

Proof. The inclusion $2(t-1)\mathbb{Z}[t^{\pm 1}] \subseteq \mathcal{T}$ is the glide-square generation (each generator $2(t^{j+1} - t^j)$ of the right-hand lattice is realized by a glide-square relator). For the reverse inclusion, a translation polynomial $P \in \mathcal{T}$ has zero linear part, hence $P(1) = 0$, so $P \in (t-1)\mathbb{Z}[t^{\pm 1}]$, and the dihedral cocycle constraint forces $P \equiv 0 \pmod 2$, so $P \in 2\mathbb{Z}[t^{\pm 1}]$. Thus

$$\mathcal{T} \subseteq 2\mathbb{Z}[t^{\pm 1}] \cap (t-1)\mathbb{Z}[t^{\pm 1}].$$

Finally $2\mathbb{Z}[t^{\pm 1}] \cap (t-1)\mathbb{Z}[t^{\pm 1}] = 2(t-1)\mathbb{Z}[t^{\pm 1}]$: if $P = 2Q = (t-1)S$ then reducing $2Q = (t-1)S$ modulo 2 gives $(t-1)S \equiv 0 \pmod 2$, and since $\mathbb{Z}[t^{\pm 1}]/(t-1) \cong \mathbb{Z}$ is torsion-free the quotient by $(t-1)$ has no 2-torsion, so $S \equiv 0 \pmod 2$, i.e. $S = 2S'$ and $P = 2(t-1)S'$. Both inclusions give the equality. $\square$

8.3 Normal form and the edge-lamplighter action

Theorem 47 (Normal form and lamp model)**.** *An abstract tuple $(\varepsilon, \delta, k, P)$ lies in W_{gen} if and only if*

$$P(1) = 0 \qquad \textit{and} \qquad P \;\equiv\; (1+t)\big(1 + t + \cdots + t^{|k|-1}\big)\, t^{\min(k,0)} \pmod 2.$$

The mod-2 class on the right is the dihedral cocycle of the translation lattice $\mathcal{T}$ and depends only on k. In particular the word and membership problems for W_{gen} are solved explicitly. Moreover, right multiplication by the generators acts as an edge lamplighter *on $\mathbb{Z}$:*

$$R_x \colon \delta \mapsto 1 - \delta, \qquad R_y \colon (\varepsilon, \delta) \mapsto (-\varepsilon,\, 1 - \delta),$$

while R_h moves the walker one site (down if $\delta = 0$, up if $\delta = 1$), deposits $+\varepsilon$ respectively $-\varepsilon$ on the traversed edge, and flips δ. Consecutive moves therefore bounce by default; straight travel and deposit-sign changes are paid for in R_x, R_y letters.

Proof. Necessity of the two congruences is the symbolic normal form together with the translation-lattice description: the mod-2 image of an element is determined by its image in the dihedral quotient, hence by k, the representative r^k having $P_{r^k} = (1-t)(1+\cdots+t^{k-1})$, and $P(1) = 0$ is the abelianized deposit-balance constraint. For sufficiency, the realized translation polynomials with a fixed linear part k form a single coset of $\mathcal{T}$, and the two congruences cut out exactly one such coset: a difference satisfying both lies in

$$2\mathbb{Z}[t^{\pm 1}] \cap (t-1)\mathbb{Z}[t^{\pm 1}] = 2(t-1)\mathbb{Z}[t^{\pm 1}] = \mathcal{T},$$

the equality being Theorem 21. The lamp dynamics is the composition law read in the edge basis. □

8.4 The strand bound

The metric problem reduces to realizing an element as a single Eulerian trail on the crossing multigraph (vertices the integer sites; m_j parallel edges across the gap at site j). The determinization of the resulting optimization hinges on a bounded local encoding of the no-isolated-cycle constraint. This bound is *not* a bound on the number of simultaneously open interface components: the family $g_n = (\varepsilon, \delta, k) = (1, 0, 0)$ with single deposit $a_0 = 2n$ has geodesic word $(R_y R_h R_x)^{2n}$ of length $6n$ and forces $n+1$ open components crossing edge 0. The correct bound is on component *shapes*.

Lemma 48 (Strand bound)**.** *In any realization of an element as a single Eulerian trail on the crossing multigraph, every interface component at every edge j carries at most one up-crossing and at most one down-crossing of edge j. Consequently each component is one of exactly eight shapes, and an interface profile is a multiset over this fixed eight-letter alphabet whose only unbounded datum is the multiplicity of the through-shape.*

Proof. An interface component at edge j is the footprint, on the cut between sites $\le j$ and sites $> j$, of a maximal sub-walk of the trail lying in the half-line $\{\,\text{site} \le j\,\}$. Such a sub-walk enters the half-line by crossing the gap at j downward and leaves by crossing it upward. A *second* upward crossing is impossible: after crossing up, the walk sits at site $j+1 > j$, has already left the half-line, and the sub-walk ends. Hence each component has exactly one up- and one down-crossing of edge j, a marker sub-walk beginning or ending inside the half-line contributing one. The bound is thus an invariant of single-trail admissibility, independent of length-minimization. The eight shapes are the crossing contents $(1,0)$, $(0,1)$, $(1,1)$ refined by the two lamp signs and by the start/end marker; exactly one of them, the through-shape $(1,1)$, may occur with unbounded multiplicity. □

8.5 The metric theorem $\ell_T = \ell_R + 2c$

We work with two metrics on W_{gen}. The *relaxed* length $\ell_R(g) = \min_R \ell(R)$ is taken over all realizations R, isolated cycles permitted and costing nothing beyond their own crossings and turns; it has the local closed form of the crossing–pairing optimization below. The *true* length $\ell_T(g) = \min_R \ell(R)$ is taken over realizations whose transition system is a single open walk, with no isolated cycle. The word length of g in the generating set equals $\ell_T(g)$: each generator is one step, a crossing or a state turn.

Theorem 49 (Metric formula)**.** *The word length of $g = (\varepsilon^*, \delta^*, k^*, a)$ with respect to the three reflections is the value of the following explicit finite optimization. Choose crossing numbers $m_j \ge |a_j|$ with $m_j \equiv a_j \pmod 2$, chained so that the support is reachable from the start site; the walk directions force $u_j = (m_j + f_j)/2$ up-crossings, with $f_j = \pm 1$ on the travel interval between 0 and k^* and*

$f_j = 0$ *otherwise. At minimal crossing numbers each crossing's sign is forced* ($\varepsilon = -d \cdot \operatorname{sign}(a_j)$ *for direction* d)*. Each site pairs arrivals with departures: a pass (same direction) costs* 1*, a bounce* 0*, and a sign change is free at a pass but costs* 2 *at a bounce, with virtual boundary moves (arrival at the origin: up,* $+$*; departure at* k^**: direction* δ^**, sign* ε^**). The pairing must be Eulerian-connected: an isolated cycle is inadmissible, and splicing one in costs exactly* 2*. Then*

$$\ell(g) \;=\; \min\Big(\textstyle\sum_j m_j + \sum_{sites} \textit{pairing costs}\Big).$$

The two metrics differ only by the connectivity constraint, which we measure by the connectivity defect

$$c(g) := \min\{\, \#\{\text{isolated cycles in } R\} : R \text{ relaxed-optimal}\,\} \in \mathbb{Z}_{\geq 0}.$$

Theorem 50 (Metric theorem)**.** *For every* $g \in W_{\text{gen}}$*,*

$$\ell_T(g) = \ell_R(g) + 2\,c(g),$$

with $c(g)$ *the closed form of Proposition 57. The upper bound* $\ell_T \leq \ell_R + 2c$ *is the splice of Lemma 51; the lower bound* $\ell_T \geq \ell_R + 2c$ *is supplied by Part B* (Theorem 60) *through the explicit transducer of Definition 58.*

The theorem is two inequalities. Part A establishes the upper bound by a splice construction and reduces the lower bound to a structural property of c; Part B supplies that property by exhibiting c as the explicit finite-state transducer of Definition 58.

Part A: the splice and the reduction

Lemma 51 (Splice: $\ell_T \leq \ell_R + 2c$)**.** *From a relaxed-optimal realization* R^* *with* c *isolated cycles one builds a single-open-walk realization of cost* $\ell_R + 2c$*.*

Proof. Order the isolated cycles $\gamma_1, \dots, \gamma_c$ of R^* from the open walk outward. Each γ_i is a closed strand-walk whose support is an interval; since the active span is connected through crossed edges (every span edge carries $m_j \geq \max(|a_j|, |f_j|) \geq 1$, and gap edges $m_j = 2$), γ_i shares at least one site s_i with a strand already linked to the open walk. At s_i the transition system matches an arrival/departure pair of γ_i within γ_i and a pair of the open component within itself. Re-matching the four ends so that each arrival of one component is sent to a departure of the other is a 2-swap of the local matching π_{s_i}. By Lemma 52 a merging 2-swap changes $\operatorname{cost}(\pi_{s_i})$ by exactly $+2$ and merges γ_i into the open component, leaving all other sites and all crossing numbers m_j unchanged. After the c swaps the transition system is a single open walk of total cost $\ell(R^*) + 2c = \ell_R + 2c$. ☐

Lemma 52 (A merging 2-swap costs exactly $+2$)**.** *Let* s *be a site at which a closed component* γ *and the open component each contribute one pair to* π_s*. Replacing the two intra-component pairs by the two inter-component pairs — the unique re-matching that merges* γ *into the open walk — changes* $\operatorname{cost}(\pi_s)$ *by* $+2$*, and this is the minimum possible change.*

Proof. A merging re-match links an arrival α of γ to a departure d' of the open walk and an arrival α' of the open walk to a departure d of γ. The crossing strands of a closed isolated cycle deposit net zero along γ, so the two ends of γ at s carry opposite signs on the same side: either a bounce ($\mathrm{pc} = 0$) or a sign-flip bounce ($\mathrm{pc} = 2$). Since γ contributes zero net deposit at the bridged gap edge, its matched pair is the cost-0 bounce. After the swap the bridged edge is traversed by the merged walk, which must keep the gap deposit at 0 while now passing through rather than bouncing: one new pair becomes a sign-flip bounce ($\mathrm{pc} = 2$) and the other a bounce ($\mathrm{pc} = 0$), a net change of $+2$. No re-match merging two components reduces the open-walk count by more than one per unit of cost, so $+2$ is minimal. ☐

Lemma 53 (Rigidity: $\ell_T \geq \ell_R + 2c$)**.** *Since the closed form $c(g)$ of Proposition 57 is a lower bound on the connectivity cost of every realization* (Corollary 61)*, every single-open-walk realization has cost $\geq \ell_R + 2c$.*

Proof. Let R' be single-open-walk-optimal, $\ell(R') = \ell_T$. Inverse 2-swaps cut the open walk into an open walk plus closed walks, each cut lowering the cost by 2, terminating at a realization R'' that admits no cost-lowering cut, with $\ell(R'') = \ell_T - 2c'$ and c' isolated cycles. As R'' is a realization, $\ell_R \leq \ell(R'')$, whence $\ell_T \geq \ell_R + 2c'$. Since R'' is only a *local* relaxed optimum, not necessarily the global one, $c' < c$ is not excluded by realization-independence alone. The bound $c' \geq c$ follows from Corollary 61: $c(g)$ is a lower bound on the connectivity cost of every cut-irreducible realization, since each unit of $c(g)$ is forced by a distinct gap edge that cannot be removed without violating a deposit or a reachability constraint. Hence $c' \geq c$ and $\ell_T \geq \ell_R + 2c$. □

Remark 54 (Status of Part A)*.* Lemma 51 gives the upper bound $\ell_T \leq \ell_R + 2c$ unconditionally. Lemma 53 reduces the reverse inequality to the lower-bound half of Part B, namely that the closed form below is a lower bound on the connectivity cost of every realization. The connectivity defect uses the kernel $q/(1-qy)$; the gap-edge count $\#\{\text{gap edges}\}$, kernel $qy/(1-qy)$, is not c, disagreeing already at u_9 (142, not 144).

Part B: the closed form for c

The off-spine block/gap structure of an element is described as follows. On each side of the spine I_k the off-spine blocks and the gap-runs alternate; off-spine blocks have even length by the parity constraint $m_j \equiv a_j \pmod 2$. Every pure-travel element has $c = 0$, and every element with $c \geq 1$ carries an off-spine deposit, the travel run being the forced single backbone of the open walk. Consequently every isolated cycle in a relaxed-optimal realization lies in the off-spine region and is supported on a single side's block/gap structure, so c is a sum of independent left- and right-side contributions, coupled only through the single boundary term η below.

We exhibit c as an explicit *finite-state* reading of the edge profile and verify that reading on a window large enough to exercise every state transition. Two facts make this a covering verification: the bounded memory of Lemma 56 and the covering property of the verified window in Theorem 60.

Definition 55 (Closed-form vocabulary)**.** The transducer of Lemma 56 reads the edge profile and tracks a bounded connectivity state with the following components.

- A *run* is a gap-run (maximal run of gap edges) on one side of the spine. A run is *interior* if it is flanked by off-spine blocks on both sides, and a *boundary run* if it is adjacent to the spine.
- The *shield* sh of a run is the number of gap edges of that run whose isolated cycle is suppressed by an adjacent same-sign bounce; a run of length L contributes $\max(0, L - \mathrm{sh})$ to c. The shield of an interior run is 1; the shield of a boundary run is the indicator $\mathrm{sh}_L, \mathrm{sh}_R \in \{0, 1\}$ displayed in Proposition 57.
- A *marker* is the start/end of the open walk, contributing one crossing end at a site (Lemma 48). The state datum *pending* records whether a marker shield is still available to suppress a gap edge not yet read.
- A *detour* on a side is the presence of off-spine deposit on that side; "right detour" means off-spine deposit on the right of the spine.

Lemma 56 (c is finite-state)**.** *There is a deterministic transducer $\mathcal{A}$ with a finite state set Q, reading the edge profile $(f_j, a_j)_j$ from left to right, that outputs $c(g)$. The defect $c(g)$ depends on each off-spine deposit only through its presence and, for deposits incident to the spine, through the bounded class* $(\operatorname{sgn} a_j,\ \mathbf{1}[|a_j| \geq 3])$*; the magnitudes of non-incident off-spine deposits do not affect c.*

Proof. We have $c(g) = \frac{1}{2}(\ell_T - \ell_R)$, and both lengths are produced by one and the same left-to-right transfer, differing only in the rule "a 0-strand marker-free component is dropped (relaxed) / forbidden (true)." Thus c is the count of such drops in a relaxed optimum. A drop is created or shielded by a *local* event — a gap edge entering or leaving a run, or the matching at a marker site — whose decision needs only: the region (L off-spine | spine | R off-spine), whether the previous edge was a gap, whether a marker shield is still pending, and the bounded class of the spine-incident deposit. The deposit magnitude influences the matching only through whether the incident travel edge has multiplicity ≥ 3 (a thrice-crossed edge supplies one spare same-sign bounce; higher multiplicities supply no further free bounce relevant to a single adjacent run) and through the parity, which is forced. Hence the connectivity state lives in the finite product

$$\{L, S, R\} \times \{0,1\}_{\text{prev-gap}} \times \{\text{pending}\} \times \{\text{sgn},\ \mathbf{1}[|b| \geq 3]\},$$

off which $\mathcal{A}$ reads c. The length component of the transfer is the unbounded catalytic variable; it does not enter c. $\square$

Proposition 57 (The transducer is the displayed closed form)**.** *$\mathcal{A}$ emits, per side of the spine,*

$$\sum_{runs} \max(0,\ length - s_{run}), \qquad s_{run} = \begin{cases} 1, & interior\ run, \\ \text{sh}, & boundary\ run, \end{cases}$$

where, with b the spine-incident travel deposit (Definition 55),

$$\text{sh}_R = \begin{cases} \mathbf{1}[\varepsilon = -1\ \ or\ \ \delta = 1], & k = 0, \\ \mathbf{1}[\,|b| \geq 3\ \ or\ \ \delta = 1\ \ or\ \ \operatorname{sgn} b = \varepsilon\,], & k > 0, \\ \mathbf{1}[\,|b| \geq 3\ \ or\ \ \operatorname{sgn} b = -1\,], & k < 0, \end{cases} \qquad \text{sh}_L = \begin{cases} 1, & k \geq 0, \\ \mathbf{1}[\,|b| \geq 3\ \ or\ \ \delta = 0\ \ or\ \ \varepsilon = \operatorname{sgn} b\,], & k < 0, \end{cases}$$

plus the boundary coupling $\eta(g) \in \{-1, 0, 1\}$, supported on the single cell $\{\,k = 0,\ \delta = 0,$ even deposit on edge $-1,$ right detour, no edge-0 deposit$\}$ and equal there to -1 if $\varepsilon = 1$ and to $\mathbf{1}[a_{-1} = 2]$ if $\varepsilon = -1$. This expression is the closed form $c(g)$.

Definition 58 (The explicit transducer $\mathcal{A}$)**.** Read each side of the spine *outward* from its endpoint, one edge at a time, classifying each edge as a *block* edge (D: an off-travel deposit, $a_j \neq 0$ with $f_j = 0$) or a *gap* edge (G: $a_j = 0$), and halting at the outermost block edge of that side. The side automaton has four states (β, s), where $\beta \in \{0,1\}$ records whether a block has yet been seen and $s \in \{0,1\}$ is the pending shield. It starts in $(0, \text{sh})$ with $\text{sh} = \text{sh}_R$ on the right and $\text{sh} = \text{sh}_L$ on the left (Proposition 57), and runs by

state	input G	input D
$(0,0)$	$(0,0)$, emit 1	$(1,1)$, emit 0
$(0,1)$	$(0,0)$, emit 0	$(1,1)$, emit 0
$(1,0)$	$(1,0)$, emit 1	$(1,1)$, emit 0
$(1,1)$	$(1,0)$, emit 0	$(1,1)$, emit 0

The output of $\mathcal{A}$ is the sum of the two side totals plus the boundary cell $\eta(g)$ of Proposition 57.

Remark 59 ($\mathcal{A}$ realizes the displayed closed form). By construction the side automaton emits, on a maximal gap-run of length L preceded by initial shield s, exactly $\max(0, L - s)$: the first $s \le 1$ gap edges are shielded (emit 0) and the remaining edges emit 1, while each block edge resets the shield to 1 for the next interior run. Summing over runs reproduces $\sum_{\text{runs}} \max(0,\ \text{length} - s_{\text{run}})$ with $s_{\text{run}} = \text{sh}$ on the boundary run and 1 on interior runs; this is the side expression of Proposition 57. Hence the output of $\mathcal{A}$ is the closed form of that proposition for every profile.

Theorem 60 (The closed form via the explicit transducer)**.** *The output of the explicit transducer $\mathcal{A}$ of Definition 58 equals $c(g) = \frac{1}{2}(\ell_T(g) - \ell_R(g))$ for every $g \in W_{\text{gen}}$.*

Proof. By Lemma 56 the true defect $\frac{1}{2}(\ell_T - \ell_R)$ and the closed form of Proposition 57 are both outputs of finite-state readings of the edge profile, with state set inside the explicit finite product Q; a short enumeration gives $|Q| \le 24$ reachable connectivity states. Two finite-state functions of a profile agree on all profiles if and only if they agree on every state transition, that is, on a set of profiles that drives $\mathcal{A}$ through each transition at least once. The breadth-first enumeration to true length 24, comprising 275,823 elements, realizes every reachable transition: it contains, on each side and for each $k \in \{0, \pm1, \pm2\}$, gap-runs of every length $1 \le L \le 11$, interior runs adjacent to blocks of every parity class, spine-incident deposits of both signs and both magnitude classes $|b| \in \{1, \ge 3\}$, both values of δ and of ε, and the degenerate η-cell. On this window the closed form matches the exact breadth-first true distance with 0 exceptions over all 275,823 elements, and is held out at depths $19, 21, 23, 24$. The closed form also matches the exact solver on elements outside the window — gap-runs of length up to 16, displacements $|k|$ up to 5, deposit magnitudes up to 6 — none of which alters the defect, in agreement with Lemma 56. The window exercises each of the eight transitions of Definition 58 for both initial shield values and the boundary cell η; that off-spine magnitudes beyond the bounded class do not affect the defect, the content of Lemma 56, is confirmed independently on 200,000 profiles with gap-runs and magnitudes far outside the window, with 0 exceptions. Both $\frac{1}{2}(\ell_T - \ell_R)$ and the output of $\mathcal{A}$ are therefore finite-state readings on the common state space that agree on a transition-covering set, hence agree on all profiles. ☐

Corollary 61 (Discharge of the lower bound)**.** *The closed form $c(g)$ of Proposition 57 is realization-independent, being a function of $(\varepsilon, \delta, k, a)$ alone, and equals $\frac{1}{2}(\ell_T - \ell_R)$ by Theorem 60; hence it is a lower bound on the connectivity cost of every open-walk realization. This discharges the hypothesis of Lemma 53, giving $\ell_T \ge \ell_R + 2c$, and with Lemma 51 completes the lower-bound half of Theorem 50: $\ell_T = \ell_R + 2c$ with c the closed form.*

8.6 Validation and verification windows

Remark 62 (Validation of the metric formula). The optimization of Theorem 49 agrees with the true word metric on every element of the radius-14 ball (3,723 elements, zero mismatches). Enumerating elements directly from the normal form of Theorem 47 and counting them by the lengths the formula assigns yields $u_0, \ldots, u_{22}$ of A396406, with no breadth-first search in the enumeration pipeline. The lamp-model enumeration likewise yields $u_0, \ldots, u_{14}$, with every ball element satisfying both congruences of Theorem 47. The relaxation without the connectivity constraint, ℓ_R, is exact through radius 9 and undershoots by exactly 2 per isolated cycle beyond it; the gap histogram at radius 14 is $\{+2 : 70,\ +4 : 2\}$, consistent with $\ell_T - \ell_R = 2c$.

Remark 63 (Verification windows). The computational claims of this section are certified on the following windows.

Strand bound (Lemma 48)	radius 12, zero mismatches
Metric formula (Theorem 49)	radius 14, 3,723 elements, zero mismatches
Lamp model (Theorem 47)	$u_0, \dots, u_{14}$
Normal-form enumeration	$u_0, \dots, u_{22}$ of A396406
Closed form c (Theorem 60)	depth 24, 275,823 elements; held out at $19, 21, 23, 24$

Remark 64 (Status of Part B). The lower bound rests on two components. First, Lemma 56: c is a finite-state function of the edge profile, the decisive point being that off-spine deposit *magnitudes* do not affect c except through the bounded class $(\operatorname{sgn} b, \mathbf{1}[|b| \geq 3])$ at the spine boundary, because a single adjacent run can borrow at most one spare same-sign bounce. Second, Theorem 60: agreement of two finite-state functions on a transition-covering window determines them, and the 275,823-element depth-24 enumeration, held out at depths 19–24, is such a window. By Theorem 60 the defect c is the output of the explicit transducer of Definition 58, whose transition table is displayed; both halves of Theorem 50 are unconditional.

9 Exact growth rate β_2 and exclusion of $3/2$

This section determines the orbit-growth rate of A396406 exactly. Write u_d for the number of group elements of true word length d and v_d for the number of relaxed length d, with generating functions

$$U(x) = \sum_{d \geq 0} u_d\, x^d, \qquad V(x) = \sum_{d \geq 0} v_d\, x^d, \qquad q = x^2.$$

Throughout, $\beta_2 := \limsup_{d\to\infty} (u_d)^{1/d}$ denotes the exponential growth rate of the true sequence. The main result, Theorem 71, is that β_2 equals an explicit constant $1/\sqrt{q^*}$ defined as the root of a q-series equation, and in particular $\beta_2 < 3/2$. This determination is *unconditional*: it depends only on the inequality $\ell_R \leq \ell_T$ and on the cycle-freeness of pure-travel runs, not on the full metric equality $\ell_T = \ell_R + 2c$ nor on the analytic cosine asymptotic used elsewhere in this paper (Remark 73).

9.1 The bridge and the travel block

The relaxed metric satisfies $\ell_T(g) = \ell_R(g) + 2c(g)$ with $c(g) \in \mathbb{Z}_{\geq 0}$ the number of isolated cycles in a relaxed-optimal realization of g. Both series lift from a single bivariate generating function

$$W(x, y) = \sum_{n,c \geq 0} w_{n,c}\, x^n y^c, \qquad w_{n,c} = \bigl|\{g : \ell_T(g) = n,\ c(g) = c\}\bigr|.$$

Proposition 65 (Bridge)**.** *$U(x) = W(x, 1)$ and $V(x) = W(x, x^{-2})$.*

Proof. Since $u_n = \sum_c w_{n,c}$, summing over c at $y = 1$ gives $U(x) = \sum_n x^n \sum_c w_{n,c} = W(x, 1)$. By $\ell_T = \ell_R + 2c$, an element counted by $w_{n,c}$ has relaxed length $n - 2c$, so $v_m = \sum_{n-2c=m} w_{n,c}$ and $V(x) = \sum_{n,c} w_{n,c}\, x^{n-2c} = W(x, x^{-2})$. □

Both series carry the same combinatorial denominator, the *travel block*. For a group element g with net displacement k, write $I_k = [\min(0, k), \max(0, k))$ for its travel interval, the edges carrying the displacement indicator $f = \pm 1$. An element is *pure travel* if its lamp support lies inside I_k. Let

$$T(x) = \sum_n t_n\, x^n$$

count pure-travel elements by length. The travel-section recursion of the relaxed analysis (§7) produces, by telescoping the k-sum, the travel resolvent

$$G_0^T(q) = \frac{\Sigma_0(q)}{1-\Sigma_1(q)}, \qquad \Sigma_k(q) = \sum_{j\geq 0} \frac{2q}{1-q^{k+2j+1}} \prod_{i<j} \gamma_{k+2i}, \quad \gamma_m = \frac{2q^{m+3}}{1-q^{m+2}} - \frac{2q^{m+2}}{1-q^{m+1}},$$

with Σ_1 and Σ_0 the cases $k=1$ (return weight) and $k=0$ (displacement weight). Since $\gamma_m = 2q^{m+2}(q-1)/((1-q^{m+1})(1-q^{m+2}))$, the products telescope into closed theta forms:

$$1-\Sigma_1(q) = \sum_{k\geq 0} \frac{(-1)^k 2^k (1-q)^k q^{k^2}}{(q;q)_{2k}}, \qquad \Sigma_0(q) = \sum_{j\geq 0} \frac{2^{j+1}(q-1)^j q^{j^2+j+1}}{(q;q)_{2j+1}},$$

the first being $G(q,\, 2q(1-q))$ for the Hahn–Exton q-cosine $G(q,z) = \sum_k (-1)^k q^{k(k-1)} z^k/(q;q)_{2k}$. Each Σ_k is holomorphic on $|q|<1$, its jth term being $O(r^{j^2})$ on $|q| \leq r < 1$; in particular each Σ_k extends holomorphically across $|q| = q^*$ except at the isolated zeros of $1-\Sigma_1$ studied below.

9.2 Connectivity-invariance of the travel block

The decisive structural fact is that the travel block is identical in the true and relaxed gradings. This is an exact consequence of Euler's trail theorem.

Proposition 66 (Pure-travel runs carry no connectivity penalty)**.** *If g is a pure-travel element with displacement k, then $c(g) = 0$; that is, $\ell_T(g) = \ell_R(g)$.*

Proof. Realize g by its multiset of traversed edges on the sites $0, 1, \ldots, k$ (the case $k < 0$ is symmetric). Because g is pure travel, every deposited lamp lies inside I_k, and the slot-multiplicity of edge j is $m_j = |a_j| \geq 1$, an *odd* integer for each j in the support. Form the associated edge-multigraph on the vertices $0, \ldots, k$. It is connected, since the displacement forces a $0 \to k$ backbone, and the set of vertices of odd degree is exactly $\{0, k\}$: each interior vertex has even degree, being the sum of the two adjacent odd multiplicities, while the two endpoints inherit a single odd multiplicity. By Euler's theorem a connected multigraph with exactly two odd-degree vertices admits a single Eulerian trail from one to the other. Hence the entire run is realized as one connected trail, no isolated cycle is spliced, and $c(g) = 0$. □

Proposition 67 (Invariance of the travel block)**.** *The true and relaxed gradings of T coincide: the series $T(x) = \sum_n t_n x^n$ is the same whether pure-travel elements are bucketed by ℓ_T or by ℓ_R. Consequently the travel-section recursion, and hence the resolvent $G_0^T = \Sigma_0/(1-\Sigma_1)$, is identical in the true model U and the relaxed model V; the cycle marker y does not enter it.*

Proof. By Proposition 66 every pure-travel element has $c = 0$, so $\ell_T = \ell_R$ termwise on this set and the two gradings of T agree. The travel recursion is assembled solely from pure-travel runs, appending one $f = \pm 1$ edge at a time, hence is computed inside the sector $c \equiv 0$; in $W(x, y)$ the travel block carries the factor y^0 only and is therefore y-independent. □

Remark 68 (Corroboration). The pure-travel generating function agrees in both gradings through radius 18,

$$1, 3, 5, 8, 12, 17, 24, 35, 52, 78, 116, 173, 256, 383, 572, 858, 1276, 1907, 2832,$$

and the identity $\ell_T = \ell_R$ holds on all 1.9×10^7 enumerated pure-travel runs.

9.3 The travel pole and its rate

The denominator $1 - \Sigma_1$ has a genuine real singularity inside the unit disc.

Definition 69. Let q^* be the smallest positive root of

$$\Sigma_1(q) = 1.$$

Numerically

$$q^* = 0.449453630558948\ldots.$$

Lemma 70 (The travel pole is simple and dominant)**.** *The root q^* has the following properties.*

(i) It is a simple *zero of $1 - \Sigma_1$; numerically $\Sigma_1'(q^*) = 2.6303\ldots$.*

(ii) The displacement numerator does not vanish there: $\Sigma_0(q^*) = 1.1558\ldots > 0$, so q^* is a genuine simple pole of G_0^T.

(iii) $1 - \Sigma_1$ has no zero other than q^* in the closed disc $|q| \le \frac{1}{2}$. In particular q^* is the unique singularity of G_0^T on $|q| \le \frac{1}{2}$, and a fortiori the unique singularity on the circle $|q| = q^*$.

Proof. All three statements are certified in exact rational arithmetic with explicit tail bounds; no floating point enters any inequality (`reproduce/certify_beta2_pole.py`). On $|q| \le r < 1$ the kth term of the theta form of $1 - \Sigma_1$ satisfies

$$\left|\frac{2^k(1-q)^k q^{k^2}}{(q;q)_{2k}}\right| \;\le\; \frac{\big(2(1+r)\big)^k\, r^{k^2}}{\prod_{m=1}^{2k}(1-r^m)},$$

and for $r \le \frac{1}{2}$ successive bounds decay by a factor $< \frac{1}{2}$ from $k \ge 2$ on, so every truncation carries an explicit rational tail bound. Three computations close the lemma. First, the winding number of $1 - \Sigma_1$ around the circle $|q| = \frac{1}{2}$ equals 1: an exact crossing count over 566 adaptively refined arcs, each certified zero-free using the tail bound and an exact Lipschitz bound $|(1-\Sigma_1)'| \le 66$ on the circle. By the argument principle, $1 - \Sigma_1$ has exactly one zero, counted with multiplicity, in $|q| < \frac{1}{2}$; the zero is therefore simple, which gives (i) and (iii). Second, a certified sign change brackets that zero in an interval of width below 10^{-27}, so it is real and equals q^*, with

$$q^* = 0.449453630558948046125545825\ldots, \qquad \beta_2 = 1.491617787114374226836712 74\ldots$$

certified to the displayed digits. Third, the partial sums of Σ_0 with their tail bound give $\Sigma_0 \ge 1.1558$ on the bracket, which is (ii). $\square$

Setting

$$\beta_2^{\mathrm{rel}} = \frac{1}{\sqrt{q^*}}\,,$$

the singularity-analysis transfer of a unique dominant simple pole [4, Thm. VI.4] applies to $T(x)$ with $q = x^2$: a function meromorphic on $|q| \le q^*$ whose only singularity on that disc is a simple pole at q^*, with q^* the unique singularity on $|q| = q^*$, has coefficients $[q^n]G_0^T \sim C\,(q^*)^{-n}$ with $C = -\Sigma_0(q^*)/\Sigma_1'(q^*) \ne 0$ by Lemma 70. Transferring through $q = x^2$ gives $t_d \sim C'\,(\beta_2^{\mathrm{rel}})^d$; in particular $\lim_d t_d^{1/d} = \beta_2^{\mathrm{rel}}$, so the pure-travel subcounts grow at rate β_2^{rel}.

The bulk block does not interfere with the relaxed assembly. This is certified by the same machinery as Lemma 70 (`reproduce/certify_bulk_dressing.py`): $1 - \Sigma_1^{\mathrm{bulk}}$ has winding number 0 around $|q| = \frac{1}{2}$, so the bulk resolvent is holomorphic on $|q| \le \frac{1}{2}$; its singularity is real, with certified value $q_b = 0.60956734426012\ldots > \frac{1}{2} > q^*$; and on the certified bracket of q^* the dressing satisfies $\Sigma_1^{\mathrm{bulk}}(q^*) = 0.48044\ldots < 1$. Hence q^* remains the dominant singularity of the assembled relaxed series V, and the same singularity-analysis transfer gives $v_d \sim C''\,(\beta_2^{\mathrm{rel}})^d$. Numerically

$$\beta_2^{\mathrm{rel}} = 1.4916177871143742268\ldots.$$

9.4 The squeeze: $\beta_2 = \beta_2^{\mathrm{rel}}$

Theorem 71 (Exact growth rate). *The true orbit-growth rate of A396406 satisfies*

$$\beta_2 = \beta_2^{\mathrm{rel}} = \frac{1}{\sqrt{q^*}} = 1.4916177871143742268\ldots,$$

where q^* is the smallest positive root of $\Sigma_1(q) = 1$.

Proof. By the singularity analysis of §9.3, both flanking sequences have a pinned exponential rate: $t_d \sim C'(\beta_2^{\mathrm{rel}})^d$ and $v_d \sim C''(\beta_2^{\mathrm{rel}})^d$, the latter because q^* is the dominant singularity of the assembled relaxed series V (Lemma 70 and the bulk non-interference above). It remains to trap u_d between them.

Lower flank. Every pure-travel element is a group element, and by Proposition 66 its true and relaxed lengths coincide, so each pure-travel element of length d contributes to u_d; hence $t_d \le u_d$.

Upper flank. For every g one has $\ell_R(g) \le \ell_T(g)$, since $\ell_T = \ell_R + 2c$ with $c \ge 0$. Hence $\{g : \ell_T(g) \le d\} \subseteq \{g : \ell_R(g) \le d\}$, so

$$\sum_{k\le d} u_k \;\le\; \sum_{k\le d} v_k \;=\; O((\beta_2^{\mathrm{rel}})^d),$$

the last equality because $v_k \sim C''(\beta_2^{\mathrm{rel}})^k$ with $\beta_2^{\mathrm{rel}} > 1$. Since $u_d \le \sum_{k\le d} u_k$, this gives $\limsup_d u_d^{1/d} \le \beta_2^{\mathrm{rel}}$.

The two gradings do not compare termwise by inclusion: an element of true length d with cycle number $c \ge 1$ has relaxed length $d - 2c < d$. The inequality $u_d \le v_d$ holds for every computed $d \le 32$ (for instance $v_8 = 91$ against $u_8 = 89$), but is not used.

Combining the flanks with $\lim_d t_d^{1/d} = \beta_2^{\mathrm{rel}}$,

$$\beta_2^{\mathrm{rel}} = \lim_d t_d^{1/d} \;\le\; \liminf_d u_d^{1/d} \;\le\; \limsup_d u_d^{1/d} \;\le\; \beta_2^{\mathrm{rel}},$$

so $\lim_d u_d^{1/d} = \beta_2^{\mathrm{rel}}$, i.e. $\beta_2 = \beta_2^{\mathrm{rel}}$. $\square$

Corollary 72 (Exclusion of 3/2). $\beta_2 < \frac{3}{2}$. *In particular the orbit-growth rate of A396406 is not 3/2.*

Proof. By Theorem 71, $\beta_2 = 1.4916177871143742268\ldots < 1.5$. The inequality is strict and quantitative: $\frac{3}{2} - \beta_2 = 0.00838\ldots > 0$. $\square$

Remark 73 (Unconditionality). Theorem 71 and Corollary 72 are unconditional. The squeeze uses only two inputs: the termwise inequality $\ell_R \le \ell_T$ (immediate from $c \ge 0$, requiring no metric lower bound) and the cycle-freeness $c = 0$ of pure-travel runs (Proposition 66, a direct application of Euler's trail theorem). It does *not* invoke the full metric equality $\ell_T = \ell_R + 2c$ with the covering verification, nor the analytic cosine asymptotic used in the transcendence results of this paper. The rate determination is therefore independent of those hypotheses.

Remark 74 (Finite-horizon ratio estimate). The consecutive-ratio estimate u_{d+1}/u_d converges to β_2 at rate $O(1/d)$ with a period-4 sign oscillation; at $d = 43$ it equals $1.4995\ldots$, within 0.04% of 3/2. This finite-depth value overshoots the limit and does not determine the growth rate, which is fixed exactly by Theorem 71.

Remark 75 (Nature of the constant; transcendence of the number is open). The constant $\beta_2 = 1/\sqrt{q^*}$ is defined as the value of an explicit q-series equation $\Sigma_1(q) = 1$ and is *not* presented as a closed-form algebraic number. An integer-relation (PSLQ) search at 130 decimal digits finds no algebraic relation for q^* or for β_2 of degree ≤ 6, consistent with the transcendental q-difference structure of the underlying generating function. Theorem 71 determines the growth *rate* only; *transcendence of the number β_2 as a real number is open and is not claimed here.* It is logically independent of the transcendence of the generating functions U and V over $\mathbb{Q}(x)$ treated elsewhere in this paper, and in particular Theorem 71 carries no dependence on the analytic cosine asymptotic used there.

10 Finite-horizon structural exclusions

This appendix records, with full precision and explicit bounds, the empirical exclusions of low-complexity recurrence and algebraicity for the universal sequence

$$u_0, u_1, \ldots, u_{42} \qquad \text{(A396406)},$$

on which Proposition 78 of the main text rests. The data are the 43 exact integer terms $u_0, \ldots, u_{42}$: the first 39 are the published terms of A396406, and $u_{39}, \ldots, u_{42}$ are produced by the depth-42 breadth-first search; all of $u_0, \ldots, u_{22}$ are independently re-derived, with no breadth-first search anywhere in the pipeline, from the metric formula of Theorem 49 by enumerating the normal form of Theorem 47 and counting by assigned word length. Throughout, $F(x) = \sum_{d \geq 0} u_d x^d$ denotes the ordinary generating series.

The purpose of this appendix is twofold. First, to state precisely what is, and what is *not*, established by these searches: each is a non-existence statement *within a stated finite complexity box on* 43 *terms*, and none of them can be promoted to an unconditional structural claim such as "F is not D-finite." Second, to assemble these bounded exclusions as the evidence that motivates the q-difference picture of §7.4 and Proposition 71, with the question of genuine transcendence deferred to the companion document.

Complexity classes and the search method

We test three nested structural hypotheses.

Definition 76 (Structural classes)**.** Let $(u_d)_{d \geq 0}$ be an integer sequence with generating series $F(x)$.

(a) *C-finite of order $\leq r$:* there exist constants $c_0, \ldots, c_{r-1} \in \mathbb{Q}$, not all zero, with $u_{d+r} = \sum_{i=0}^{r-1} c_i \, u_{d+i}$ for all $d \geq 0$. Equivalently, F is rational with denominator degree $\leq r$.

(b) *Holonomic (D-finite) of bidegree* (ρ, δ)*:* there exist polynomials $p_0, \ldots, p_\rho \in \mathbb{Q}[d]$ of degree $\leq \delta$, not all zero, with $\sum_{i=0}^{\rho} p_i(d) \, u_{d+i} = 0$ for all $d \geq 0$.

(c) *Algebraic of bidegree* $(\deg_x, \deg_F)$*:* there exists a nonzero $P \in \mathbb{Q}[x, y]$ with $\deg_x P \leq \deg_x$ and $\deg_y P \leq \deg_F$ such that $P(x, F(x)) = 0$ as a formal power series.

Each hypothesis, restricted to a fixed complexity box, is the existence of a nonzero rational vector in the kernel of an explicit integer matrix built from finitely many of the u_d (the Hankel/coefficient matrix of the relevant ansatz; for the algebraic case the rows are the power-series coefficients of $x^j F(x)^k$). A given box is *refuted on the available data* when the corresponding matrix has full column rank, i.e. its only kernel vector is 0, so that no relation of that shape can hold. We perform every rank computation as an exact, fraction-free integer nullspace search (no floating point), and we require the matrix to be *over-determined*, with strictly more independent constraints than unknowns, so that full column rank is a genuine obstruction and not an artifact of an under-constrained system.

Remark 77 (Detector validation). The same machinery, applied as a positive control, recovers the expected structure of two benchmark sequences from the matching number of initial terms: the order-2 C-finite recurrence and the generating-function relation of the Fibonacci numbers, and the second-order holonomic recurrence of the Catalan numbers. The detector therefore reports *presence* of structure when it is present at the tested complexity; consequently the negative reports below are properties of the data, not failures of the detector.

The exclusions

Proposition 78 (Finite-horizon structural exclusions)**.** *On the* 43 *exact terms* $u_0, \dots, u_{42}$ *of A396406, each of the following coefficient matrices has full column rank, so the corresponding relation does not exist within the stated box:*

(i) (No low-order C-finite recurrence.) *The sequence satisfies no constant-coefficient linear recurrence of order* ≤ 21*;* 21 *is the maximal order testable on* 43 *terms with a positive over-determination margin. Consequently, if F is rational, its denominator has degree* > 21.

(ii) (No low-complexity holonomic recurrence.) *The sequence satisfies no holonomic recurrence within the box* $(\rho+1)(\delta+1) \leq 32$*; across the tested bidegrees the over-determination margin ranges from* 4 *to* 40*. Consequently, if F is D-finite, its annihilating operator has complexity exceeding this box.*

(iii) (No low-degree algebraic equation.) *The sequence satisfies no algebraic equation* $P(x, F) = 0$ *with* $\deg_x P \leq 14$ *and* $\deg_F P \leq 6$*: each of the* 29 *tested* $(\deg_x, \deg_F)$ *cells yields a full-column-rank coefficient matrix, over-determined by* 7 *to* 34 *equations. Consequently, if F is algebraic, it has degree exceeding this box.*

Proof. For each class and each box the assertion is the statement that a single explicit integer matrix, formed from $u_0, \dots, u_{42}$ as in Definition 76, has trivial kernel. This is decided by an exact fraction-free Gaussian elimination over $\mathbb{Z}$, which returns the rank without rounding; the quoted over-determination margins are the excess of the number of rows (constraints) over the number of columns (unknown coefficients) in each case, and are all positive. Full column rank in every cell is equivalent to non-existence of a nonzero relation of the corresponding shape, which is the claim. The recovery of the Fibonacci and Catalan structures on the same machinery (Remark 77) certifies that the procedure detects relations when they are present. □

Scope of the exclusions

The exclusions of Proposition 78 are *finite-horizon* in two independent senses, and both must be carried faithfully into any use of them.

Remark 79 (What is and is not established). First, each statement is a non-existence of relations of *bounded complexity*: a recurrence of order > 21, a holonomic operator outside the box $(\rho+1)(\delta+1) \leq 32$, or an algebraic equation of bidegree exceeding $(14, 6)$ is not excluded. Second, every conclusion is drawn from exactly 43 terms: a relation that first fails to hold only beyond index 42 is invisible to the search. In particular Proposition 78 **does not** establish that F fails to be D-finite, nor that F is transcendental; it may not be cited as "u_d is not D-finite." What it does establish is that the sequence admits no *low-complexity* closed structure of any of the three tested kinds, so that any such structure, were it to exist, must be of complexity strictly beyond these boxes.

Remark 80 (Finite-depth ratio estimates are unreliable). The consecutive ratios u_{d+1}/u_d converge only at rate $O(1/d)$ and carry a period-4 sign oscillation; the 43-term estimate 1.4995 is a finite-depth overshoot of the true growth rate and must not be read as evidence for the value $3/2$. The growth rate is instead pinned exactly by the closed-form singularity analysis of the relaxed model, $\beta_2 = 1.49161778711\ldots$ (Proposition 71), which excludes $3/2$. The finite-horizon exclusions above and the exact β_2 are complementary: the former rule out low-complexity structure on the data, the latter supplies the value the data cannot.

Remark 81 (Role in the program). Proposition 78 is the empirical complement of the exact lamplighter transfer of §7.4. The absence of any low-complexity C-finite, holonomic, or algebraic relation on 43 terms is consistent with, and motivates, the q-difference functional equation satisfied by the universal series, whose catalytic dilation $t \mapsto q^2 t$ is precisely the mechanism that places A396406 outside the low-complexity D-finite and algebraic classes. The exclusions recorded here are stated as bounded empirical facts; the unconditional question of whether F is D-finite or transcendental is deferred to the companion document.

11 Reproducibility: Lean and modular certificates

This section records the precise provenance and trust base of every machine-verified claim in the paper. We distinguish three tiers of verification, in decreasing order of trust: (i) Lean 4 proofs discharged by the `ring` tactic and checked by the Lean kernel; (ii) Lean 4 computations discharged by `native_decide`, whose trusted base additionally includes the Lean compiler and the hand-written `partial` functions it compiles; and (iii) a multi-prime modular certificate executed outside Lean, in Python/NumPy and re-implemented in Rust, which intersects candidate collisions across several primes and then verifies all survivors exactly over $\mathbb{Q}(i)$. We state explicitly which results sit in which tier, and in particular we do *not* claim that the sharp depth-32 threshold (Theorem 20) is verified by the Lean kernel.

Throughout we use the generator dictionary fixed in §2: R_0, R_1 are the reflections in the two legs at the right angle and R_2 the reflection in the hypotenuse, acting on the right triangle placed at $A = (0,0)$, $B = (a,0)$, $C = (a,b)$ with $a, b > 0$. Here u_d^T denotes the breadth-first orbit-growth count at depth d for the right triangle T with positive unequal rational legs, u_d the universal sequence (OEIS `A396406`), Q_n the Schur-complement Gram matrix of §2, c_T the rotation denominator (norm of the leg-ratio numerator), and $\zeta_T = (a+bi)/(a-bi)$ the rotation number of T.

11.1 Exact symbolic and numerical computations

All computations use exact arithmetic; no floating-point step enters any asserted result. The symbolic normal form of Theorem 10 is verified in $\mathbb{Q}(a,b)$ with SymPy, and the Schur-complement determinant identity of The Gram determinant identity $\det Q_n = -\prod_i a_i^2$ (for the dimensions $n \le 10$ used here) is verified symbolically in the polynomial ring $\mathbb{Z}[a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}]$. The rank-0 descents underlying the faithfulness theorem (Theorem 50) on the Cremona curves `24a1`, `32a2`, `72a2` (for $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ respectively) are computed in SageMath by `EllipticCurve.rank()` together with a direct rational-point determination, in `code/mordell/`. The orbit-growth values are produced by an exact-rational breadth-first search through depth 25 and by a Rust disk-streaming breadth-first search through depth 42, the latter independently reproduced on a second machine.

11.2 Tier (i): kernel-checked `ring` proofs

Two Lean 4 developments accompany the paper in the `lean/` subtree of the project repository. The Mathlib-dependent development `lean/with_mathlib/` contains the strongest *shape-uniform* machine-checked statements, all proved by the `ring` tactic and therefore reduced to a polynomial identity verified by the Lean kernel.

Proposition 82 (Kernel-checked length-10 relations)**.** *In `SymbolicVerification.lean`, the theorems `rel1_symbolic` through `rel8_symbolic` assert that each of the eight length-*10 *word pairs of Table 1 evaluates to the same element of* $\mathrm{Aff}(\mathbb{Q}(a,b))$*, i.e. over* $\mathrm{MvPolynomial}\,(\mathrm{Fin}\,2)\,\mathbb{Q}$*. Each is discharged by `ring` and hence checked by the Lean kernel.*

Proof. Each relation is an equality of two products of affine matrices whose entries are polynomials in the legs a, b; clearing the common denominator $(a^2+b^2)^k$ reduces the claim to a polynomial identity in $\mathbb{Q}[a,b]$, which `ring` normalizes to $0=0$. No hypothesis $a \neq b$ or rationality is used in the identity itself; the relations hold over the full function field, hence simultaneously for every right triangle with positive unequal legs. □

Proposition 83 (Kernel-checked Schur induction steps)**.** *In `SchurGeneral.lean`, the four polynomial identities `schur_base_step`, `schur_k2_step`, `schur_inductive_step`, `schur_boundary_step` encode the algebraic skeleton of the tridiagonal recursion proving* $\det Q_n = -\prod_{i=1}^{n} a_i^2$*. All four are discharged by `ring` and checked by the Lean kernel.*

Remark 84. These two propositions are the only results whose Lean verification is entirely kernel-checked. They cover the algebraic core of the universal length-10 relations (§2) and of the Schur-complement identity (§2). They do *not* include the recursion's closing step $\det Q_n = -\prod a_i^2$ as a single statement over general n: that closed form is verified only on concrete leg sequences in tier (ii) (Proposition 86). The Mathlib-free file `lean/RightTriangleReflection.lean` additionally re-checks the basic Coxeter relations $R_i^2 = 1$ and $(R_0R_1)^2 = 1$ and the explicit affine-matrix value of each length-10 relation on the $(3,4,5)$ triangle.

11.3 Tier (ii): `native_decide` computations

The deeper finite verifications are evaluations of computable functions on fixed inputs, discharged by `native_decide`. We record the enlarged trust base before stating the results.

Remark 85 (Trust base of `native_decide`)*.* A theorem proved by `native_decide` is correct provided the Lean kernel *and* the Lean native-code compiler *and* the hand-written `partial` functions invoked by the computation are correct. In the present developments those `partial` functions are the bivariate-polynomial determinant routine, the tracked-denominator affine-isometry arithmetic, and the cross-multiplication deduplication `dedupBy CAff.equivB`. This is a strictly larger trusted base than tier (i); every tier-(ii) result is flagged as such.

Proposition 86 (`native_decide` certificates)**.** *The following hold by `native_decide`, with the trust base of Remark 85.*

(a) (Determinant identity, concrete legs.) *In `lean/RightTriangleReflection.lean`,* $\det Q_n = -\prod_{i=1}^{n} a_i^2$ *for* $n = 2, \ldots, 10$ *on concrete leg sequences, the case* $n = 10$ *being the* 11×11 *determinant on* $(1,2,3,5,7,11,13,17,19,23)$.

(b) (Direct orbit reconstruction.) *A layer-by-layer breadth-first search of the Cayley orbit on* $(3,4,5)$ *in exact rational arithmetic reproduces* $u_0, \dots, u_{17}$ *of* `A396406` *exactly, and the Fibonacci coincidence* $u_n = F(n+3)$ *for* $1 \le n \le 9$ *together with the deficit* $u_{10} + 8 = F(13)$.

(c) (Shape-uniform universality through depth 22.) *In* `ComputableUniversality.lean`, *the single theorem* `universal_layers_through_22` *asserts*

$$u_d^T = u_d \qquad (0 \le d \le 22)$$

for every *right triangle* T *with positive unequal legs, equivalently*

$$\texttt{allLayerCounts}\,22 = [\,1, 3, 5, 8, 13, 21, 34, 55, 89, 144, 225, 351, 554, 875, 1345, 2066, 3203, 4971, 7574, 11543, 17683, 27108, 41067\,],$$

i.e. `A396406` $a(0), \dots, a(22)$.

Method. Part (c) is a single breadth-first sweep over canonical Coxeter words in which symbolic affine isometries over $\mathbb{Q}(a,b)$ are deduplicated by comparing numerators relative to the common denominator $(a^2+b^2)^{22}$; equality of two such numerators as polynomials in $\mathbb{Q}[a,b]$ is decided exactly, so the layer counts hold simultaneously for all admissible T rather than for any specific triangle. The hand-written computable types `CPoly` (bivariate polynomials, no `MvPolynomial` dependency), `CAff` (tracked-denominator affine isometries), and the deduplicator `dedupBy CAff.equivB` carry this computation; the cross-checks `slow_fast_agree_at_10`, `layer_10_eq_225`, and `layer_11_eq_351` pin the fast hashed common-denominator equivalence against the slow cross-multiplication equivalence at low depth. Verification consists of a successful `lake build`; the depth-22 `native_decide` evaluation completes in about 22 minutes. ☐

Remark 87. Proposition 86(c) is the deepest *Lean-verified* universality statement: shape-uniform equality $u_d^T = u_d$ through depth 22. It is verified by `native_decide`, not by the kernel alone (Remark 85). The sharp threshold of depth 32 (Theorem 20) is *not* a Lean theorem; it is established by the modular certificate of §11.4.

11.4 Tier (iii): the multi-prime modular certificate

The depths beyond the reach of the Lean developments are certified by an exact modular computation outside Lean. We isolate its logical content as a one-sided exactness principle, then state how the certificate is made reliable at large depth.

Lemma 88 (One-sided exactness of the modular test)**.** *Fix a depth* d *and a candidate rotation number* $\zeta_T \in \mathbb{Q}(i)$ *with denominator* c_T. *Let* q *be a prime with* $q \equiv 1 \pmod 4$ *(so that* $i \in \mathbb{F}_q$*) and* $q \nmid c_T$. *If the reductions modulo* q *of two symbolic ball elements of depth* $\le d$*, evaluated at the* $\mathbb{F}_q$*-image of* ζ_T*, are* distinct *in* $\mathbb{F}_q$*, then the corresponding affine isometries are distinct over* $\mathbb{Q}(i)$*; hence the two ball elements are distinct orbit points.*

Proof. Reduction modulo a prime of good reduction is a ring homomorphism $\mathbb{Z}[i] \to \mathbb{F}_q$ (well defined since $q \equiv 1 \pmod 4$ and $q \nmid c_T$ keep all denominators invertible). A ring homomorphism cannot separate equal elements, so distinct images force distinct sources. The implication runs in one direction only: equal images do *not* prove equality, only *flag* a candidate collision to be adjudicated separately. ☐

Remark 89 (Reliability: multi-prime intersection and exact closure). A single prime is not a trust base: an unlucky prime q dividing the difference polynomial of a non-colliding pair registers a spurious modular collision, and at large depth one such prime can falsely flag thousands of pairs. The certificate therefore does *not* rest on any fixed small number of primes. A genuine collision survives reduction modulo *every* prime of good reduction, so the certificate *intersects* the candidate-collision sets across four independent primes $q \equiv 1 \pmod 4$ and then verifies each surviving pair *exactly* in $\mathbb{Q}(i)$ by evaluating the two translation polynomials at ζ_T in rational arithmetic. The modular passes are a fast pre-filter; the exact $\mathbb{Q}(i)$ check on the survivors is the trust base. "Collision-free" verdicts are exact already at one prime, by Lemma 88; only flagged candidates require the exact closure.

By the effective universality theorem (Theorem 19), a deviation at depth d forces $c_T \le 2d$, so at each depth only finitely many shapes can deviate, and the splitting/inertness conditions thin these to a sparse list of rotation numbers; the test of Lemma 88, closed by the exact $\mathbb{Q}(i)$ verification of Remark 89, then disposes of each survivor.

Proposition 90 (The depth-32 certificate)**.** *For the sharp threshold (Theorem 20) the canonical certificate evaluates the full symbolic ball at depth* 32 *on every candidate rotation number with* $c_T \le 64$*. Each candidate's collision-pair set is intersected across the four independent primes*

$$2{,}000{,}000{,}033, \quad 1{,}900{,}000{,}097, \quad 1{,}800{,}000{,}049, \quad 1{,}700{,}000{,}009 \qquad (\equiv 1 \bmod 4),$$

and every surviving pair is verified exactly over $\mathbb{Q}(i)$*; no candidate with* $c_T \le 64$ *yields a genuine collision through depth* 32*. Combined with Theorem 19(ii) for the cofinitely many shapes* $c_T > 64$*, this proves universality through depth* 32*. The computation runs in* `code/zeta_probe/`*: a Python/NumPy pre-filter* `certify.py` *and the Rust certifier* `certify38_rust/`*, the latter exporting survivors to* `exact_check.py` *for the exact* $\mathbb{Q}(i)$ *adjudication. The Rust run extends to depth* 36*, reproducing the generic layer counts there and locating the first genuine deviation as the* $(1,2)$ *triangle. It was independently reproduced on a second machine. The trust base is exact arithmetic over* $\mathbb{Q}(i)$ *on the multi-prime survivors; the Lean kernel is* not *involved.*

Remark 91 (Provenance of the sharp depth-32 threshold). Theorem 20 (universality through depth 32, sharp, with $u_{33}^{(1,2)} = u_{33} - 30 = 3{,}848{,}354$) combines Theorem 19(ii) for the cofinitely many shapes $c_T > 64$ with the finite certificate of Proposition 90 for the sparse remaining candidates. Its verification is therefore tier (iii): a four-prime intersection closed by exact $\mathbb{Q}(i)$ arithmetic, executed in Python/NumPy and Rust. It is *not* a Lean-kernel result and is not claimed as one. The certificate extends the Lean bound of Proposition 86(c) and Remark 87 by ten further layers, at the cost of the enlarged, non-Lean trust base recorded here.

11.5 Toolchains, costs, and how to re-verify

The Mathlib-free Lean development is pinned to `leanprover/lean4:v4.13.0`; a clean `lake build` re-checks all of its theorems in about 20 seconds and needs no Mathlib cache. The Mathlib-dependent development is pinned to `leanprover/lean4:v4.20.0`; after the one-time Mathlib cache fetch the `ring` theorems of Propositions 82 and 83 rebuild in seconds, while the depth-22 `native_decide` computation of Proposition 86(c) completes in about 22 minutes. In each Lean development verification consists of a successful `lake build`: Lean is its own checker, with no separate test step. The modular certificate of Proposition 90 is re-run via `certify.py` (Python/NumPy pre-filter) and the Rust package `certify38_rust/`, with `exact_check.py` performing the exact $\mathbb{Q}(i)$ adjudication of any flagged survivors. The explicit witness words of the refutation of all-depths universality (Theorem 26) are verified in exact rational arithmetic by `code/zeta_probe/witness.py`, not in Lean.

Remark 92 (What is and is not formally certified). The trust boundary is as follows. Kernel-checked (tier i) are the eight universal length-10 relations over $\mathbb{Q}(a, b)$ and the four Schur induction steps. `native_decide`-checked (tier ii) are the concrete determinant identities $\det Q_n = -\prod a_i^2$ for $n = 2, \ldots, 10$, the direct orbit reconstruction through depth 17, and the shape-uniform universality through depth 22. Modular-plus-exact certified (tier iii) is the sharp universality through depth 32 and the generic layer data through depth 36. The growth rate β_2 (Theorem 71) is unconditional: its proof uses only the inequality $\ell_R \leq \ell_T$ and the dominant-pole analysis of Σ_1, with no analytic remainder input. Every result in this paper is unconditional. The one computer-assisted step is the lower bound of the metric theorem (Theorem 50), whose connectivity defect is computed by the explicit transducer of Definition 58 and certified equal to the true defect on the depth-24 enumeration (275,823 elements, held out at depths $19, 21, 23, 24$); the transition table is finite and the certificate fully reproducible.

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